

THE ROLE OF INTERPERSONAL UNCERTAINTY IN PROSOCIAL BEHAVIOR

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Abstract

In prosocial decisions, decision-makers face *interpersonal uncertainty*: uncertainty about how their choices affect others' utility. We study how this uncertainty shapes behavior jointly with social preferences. We propose a two-part mechanism: individuals are averse to interpersonal uncertainty, and they perceive different levels of uncertainty for different people. We experimentally test this mechanism in three ways: by exogenously varying interpersonal uncertainty, by eliciting perceived uncertainty directly, and by comparing standard allocation decisions with decisions that remove social consequences while preserving uncertainty. We find that interpersonal uncertainty is measurable, quantitatively important, and predicts choices across individuals, decisions, and social interventions.

Keywords: prosocial behavior, decision-making under uncertainty, ingroup favoritism, self-favoring behavior

JEL Classification: C91, D01, D91

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Research transparency: All studies were preregistered ([#159530](#), [#161634](#), [#174673](#), [#159768](#), [#162610](#), [#180119](#), [#209617](#)). See Appendix F for details. The instructions of all studies are available at [OSF](#).

1 Introduction

Social preferences imply that individuals care about the well-being of others. To act on such preferences, however, decision-makers must form beliefs about how their choices affect others' utility. This is difficult because individuals directly experience only their own utility, not the utility of others. Prosocial decisions therefore involve uncertainty about how much utility others gain or lose from a given allocation. We refer to this as *interpersonal uncertainty*.

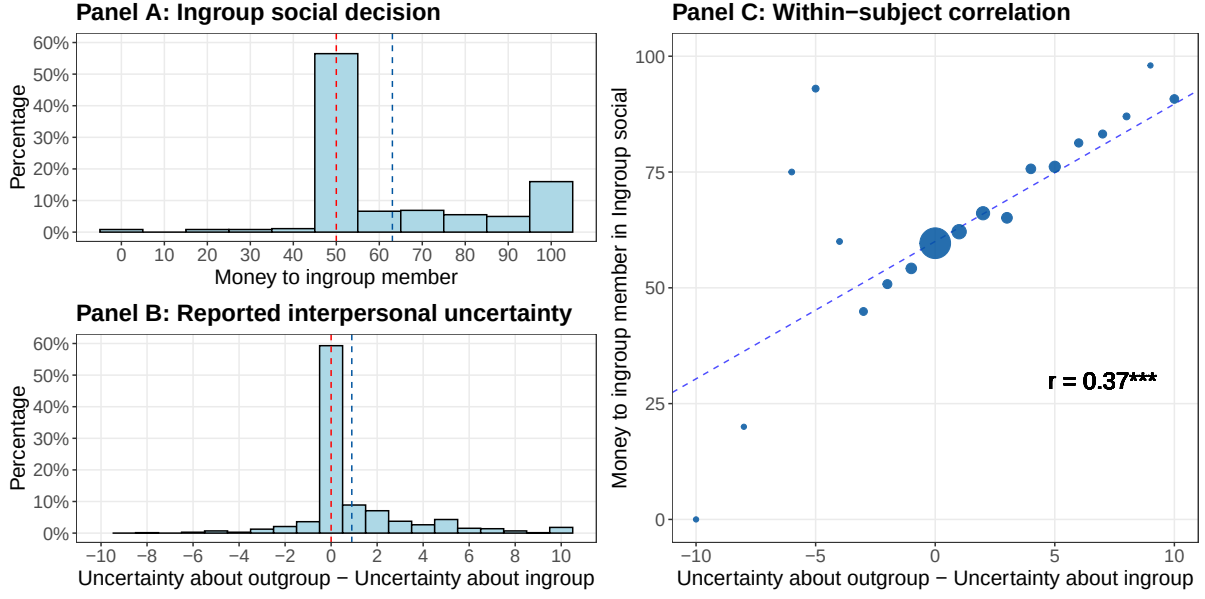
In this paper, we show theoretically and demonstrate empirically how interpersonal uncertainty shapes behavior across key paradigms in the social preference literature. We hypothesize a two-part mechanism: (i) individuals are *averse* to interpersonal uncertainty, and (ii) they perceive *different* levels of interpersonal uncertainty for different people. For instance, in the context of ingroup versus outgroup decisions - our primary application - individuals may perceive greater uncertainty about outgroup members than about ingroup members, since they are more likely to share past experiences, economic status, and tastes with their ingroup. The contribution of the paper is to show that this uncertainty is measurable, behaviorally relevant, and quantitatively important alongside social preferences.

Figure 1 provides motivating evidence for our mechanism for ingroup versus outgroup decisions. Using our experimental data, in Panel A we replicate the key behavioral pattern typically found for these decisions: when allocating a sum of money between an ingroup member – who shares a social group with the decision-maker – and an outgroup member, the ingroup member receives more money. This pattern has been documented many times in the literature¹ and shown to underlie many important processes, for instance hiring (Giuliano, Levine, and Leonard, 2009), support for policies (Enke, Rodríguez-Padilla, and Zimmermann, 2023), and the provision of public resources (Burgess et al., 2015). Panel B provides evidence for the belief component of our mechanism: subjects report being more uncertain about the utility of outgroup members relative to ingroup members. Panel C links these two patterns at the individual level: higher reported uncertainty about the outgroup relative to the ingroup is associated with higher giving to the ingroup. This correlation is consistent with aversion to interpersonal uncertainty reinforcing ingroup favoritism.

While the correlation in Panel C motivates our proposed mechanism, to establish the mechanism, we need to separate social preferences and social beliefs, which are bundled together in standard allocation choices. Our conceptual framework makes this explicit. In the model, a decision-maker holds beliefs about how much each recipient values the allocation. An allocation choice can therefore reflect three distinct components: how much the decision-maker cares about each recipient's utility, what she believes each recipient's average valuation is, and how uncertain she is about those valuations. The framework shows that, under risk aversion, higher uncertainty about one recipient's valuation can shift alloca-

¹For instance, Shayo (2020) lists 65 studies documenting ingroup favoritism across different groups, domains, and methods. Balliet, Wu, and De Dreu (2014) analyze 212 studies from 77 articles.

Figure 1: Ingroup allocation decisions, reported interpersonal uncertainty and their association



Notes: **Panel A:** Histogram showing the allocation decisions of subjects who were asked to allocate a fixed sum between an ingroup member and an outgroup member. **Panel B:** Histogram showing the difference in reported interpersonal uncertainty between ingroup and outgroup members. Higher numbers indicate that subjects are less certain about the utility of the outgroup member relative to that of the ingroup member. **Panel C:** Binned scatter plot showing the relationship between reported interpersonal uncertainty and allocation decisions. We divide the interpersonal uncertainty measure into 20 bins and plot the average allocation within each bin. In Panels A and B, the red line denotes the even split, and the blue line denotes the average allocation. In Panel C, the blue line indicates the linear regression fit, and the correlation coefficient is displayed ($***p < 0.01$). $n = 720$ decisions from 240 subjects, pooled across three social groups and two environments (*Effort* and *Gift card*; see Section 3.4).

tions away from that recipient even when expected valuations and social weights are held fixed. Interpersonal uncertainty can therefore reinforce ingroup favoritism. The empirical challenge is therefore to separately identify the importance of social preferences and beliefs in prosocial decisions. We use four complementary experimental paradigms to overcome this challenge and provide evidence for the two-part mechanism of interpersonal uncertainty.

Measuring attitudes towards interpersonal uncertainty. Our first paradigm asks subjects to allocate money between two *synthetic groups*, each consisting of two recipients. Subjects observe the distribution of recipients' valuations for bonus money in each synthetic group. We vary the spread of this distribution to vary interpersonal uncertainty. We also vary which group consists of ingroup or outgroup members, and whether that group information is provided at all. This allows us to identify subjects' aversion to interpersonal uncertainty independently of ingroup preferences. In the absence of group information, subjects allocate significantly more of their endowment to the group with lower valuation variance, revealing aversion to interpersonal uncertainty. When group identity and valuation uncertainty are both present, allocations shift in the direction predicted by the framework: subjects allocate more to the ingroup when the outgroup is more uncertain, and less to the ingroup when

the ingroup is more uncertain. Moreover, when given the choice of which information to acquire before a subsequent allocation decision, 81% of subjects choose to acquire recipients' valuation information rather than recipients' ingroup/outgroup status, indicating that such information is actively demanded.

Measuring perceived uncertainty. Our second paradigm directly measures perceived interpersonal uncertainty. Across elicitation contexts, reward media, and social groups, subjects report significantly greater uncertainty about outgroup members' valuations than ingroup members' valuations. At the same time, they do not report systematic differences in average valuations. Thus, the belief difference is not that outgroup members are perceived to value the allocation less on average, but that subjects are less certain about how they value it.

Calibrating uncertainty and attitudes jointly. Our third paradigm combines the first two paradigms to quantify the importance of our novel mechanism. In a calibration experiment, we estimate how each subject maps objective valuation spreads into perceived uncertainty, and subjects separately report their uncertainty about ingroup and outgroup members. Subjects then allocate money between two synthetic groups of recipients whose objective valuation spreads reproduce each subject's own reported ingroup-outgroup uncertainty difference. This allows us to quantify how much observed ingroup favoritism is accounted for by interpersonal uncertainty and subjects' response to it. We estimate that interpersonal uncertainty accounts for at least 38% of observed ingroup favoritism.

Social and non-social decisions. Finally, as a fourth paradigm, we correlate social and non-social decisions as complementary evidence. In the social decisions, subjects make standard allocation choices with social consequences for ingroup and outgroup recipients. In the non-social decisions, we remove all social consequences to create an individual payoff-maximizing problem, where the payoff depends solely on the subjects' beliefs about others' valuations. We find that behavior in the non-social decisions is strongly correlated with behavior in the social decisions at the individual level. Additional robustness treatments show that subjects respond systematically to changes in the non-social incentives, which speaks against confusion, inattention or framing as explanations for the pattern. These results support the interpretation that uncertainty-related forces operate alongside social preferences in shaping allocation behavior.

Taken together, these results make two key contributions. First, they refine what can be inferred from standard social-preference measures. Prosocial behavior reflects not only how much decision-makers care about recipients; it also reflects what decision-makers believe about recipients' valuations and how uncertain they are about those beliefs. Accounting for interpersonal uncertainty therefore helps measure social preferences more precisely and we provide the methodological tools to do so.

Second, our findings identify an additional lever for changing prosocial behavior. Changes in social preferences may require changes in identity, norms, concern, or social distance. By contrast, interpersonal uncertainty can be changed through information about recipients' needs, circumstances, or valuations. This distinction helps explain why information-rich interventions, such as identifiable-victim and contact interventions, can affect generosity.

We examine this implication directly by studying two well-known interventions: the *identifiable victim effect* and *contact interventions*. We find that both interventions increase generosity and reduce reported uncertainty about the recipient's valuation or experience. Next, using the framework of Kwon and Roth (2026), we test the sharp null that the increase in generosity in these interventions is driven *completely* by the reduction in interpersonal uncertainty. We fail to reject the sharp null for the *identifiable victim effect*, underscoring the principal role uncertainty-reduction plays in the *identifiable victim intervention*. In comparison, the test results suggest that uncertainty-reduction plays only a supporting role in the *contact intervention*, alongside other channels. Overall, our results suggest that interpersonal uncertainty serves as a unifying mechanism that, alongside social preferences, helps explain prosocial behavior across different paradigms and interventions.

Building on the evidence that interpersonal uncertainty shapes ingroup-outgroup behavior, we then investigate its importance in shaping self versus other behavior. Our framework naturally extends to this context: DMs may naturally know their own preferences better than they know others' preferences. We use our *non-social* approach to create a *Self social* and a *Self non-social* decision scenario. While the former is a standard dictator game where a DM's decision has consequences for a recipient, the *Self non-social* decision again is an individual payoff-maximizing problem with no consequences for others. We find "selfish-looking choices" in the *non-social*, and a high correlation between the *Self social* and the *Self non-social* decisions. These results suggest that responses to interpersonal uncertainty also shape self versus other decisions.

Related literature. We demonstrate how interpersonal uncertainty influences patterns of prosocial behavior that have been documented across two different strands of the literature. First, a large literature finds evidence that people behave more prosocially towards ingroup members, a finding that is robust across different groups, domains, and methods (such behavior has been labeled ingroup favoritism, parochial altruism or moral universalism, see Charness and Chen, 2020; Shayo, 2020; Enke, 2024, for recent overviews).² Second, many studies have documented that in *self versus other* decisions, most people behave prosocially but tend to make choices that favor themselves more than others (see Capraro, Halpern, and Perc, 2024; Fehr and Charness, 2025, for overviews). We propose a unified,

²See Iyengar et al. (2019) and Böhm, Rusch, and Baron (2020) for a review of the recent literature on ingroup favoritism in political science and psychology, respectively.

belief-based mechanism that contributes to both patterns of prosocial behavior.³ Our results show that due to the inherent presence of uncertainty, observed prosocial behavior cannot be interpreted solely as expressions of social preferences even in standard elicitation tasks. In particular, our results imply that these tasks overestimate the extent of ingroup preferences (or taste-based discrimination) and underestimate the degree of altruism in dictator games. We provide a methodology to separately identify and quantify the roles of beliefs and preferences in driving prosocial behavior.

Our paper also sheds new light on the empathy literature. While prior work shows that perceived similarity or *relatability* increases prosocial behavior (Andries et al., 2025), our framework suggests that relatability may enhance generosity not only by increasing other-regarding concerns, but also by boosting confidence in one’s ability to infer and predict others’ well-being. In this way, *empathic confidence* – as measured by interpersonal uncertainty – provides a new bridge between the role of relatability in prosocial behavior and the metacognitive processes that govern prosocial decisions.

With our subjective uncertainty-based explanation of prosocial behavior, we relate to a recent literature that explains a range of behavioral patterns through people’s cognitive response to (subjective) uncertainty. Enke and Graeber (2023) investigate how people’s uncertainty over the optimal decision influences choice under risk, belief formation, and forecasts. In the domain of intertemporal decisions, a series of theoretical studies show that risk and time preferences closely intertwine when DMs are uncertain about future consumption (Sozou, 1998; Dasgupta and Maskin, 2005; Halevy, 2008; Chakraborty, Halevy, and Saito, 2020) or future preferences (Amador, Werning, and Angeletos, 2006; Chakraborty, 2021). While this literature focuses on characterizing a logical equivalence between subjective uncertainty and risk or time preference patterns, we study the connection between subjective uncertainty and prosocial behavior both theoretically and empirically. In particular, our calibration experiment allows us to quantify the extent to which subjective uncertainty in the form of interpersonal uncertainty drives prosocial behavior. We thus differ from papers investigating prosocial behavior under experimenter-induced *objective risk* over consequences to study ex-post versus ex-ante fairness (e.g., Brock, Lange, and Ozbay, 2013), the use of risk to act selfishly (Exley, 2016), or risk-induced morality (Chen and Zhong, 2025).⁴

³Previous belief-based explanations of prosocial behavior have been mainly applied to strategic interactions, such as trust or reciprocity (Berg, Dickhaut, and McCabe, 1995; Fehr and Gächter, 2000).

⁴Further, Cappelen et al. (2022) and Cappelen, De Haan, and Tungodden (2024) study redistribution decisions when the source of inequality is uncertain. Moreover, Kappes et al. (2018) vary uncertainty about the wealth level of recipients, finding evidence against the hypothesis that people exploit such uncertainty to license more selfishness.

2 Conceptual framework

Building on our central premise that decision-makers (DMs) experience *interpersonal uncertainty* when making choices that affect others, we introduce a simple "as if" model to formalize this idea. In this model, a DM allocates \$100 between two recipients and evaluates each outcome using a function U that combines the recipients' utilities symmetrically.⁵ We use this symmetric benchmark to isolate the effect of interpersonal uncertainty from preference-based reasons to favor one recipient over the other. The DM represents the interpersonal uncertainty through two probabilistic priors, one for each recipient's potential utilities. Next, we formally define what it means to experience higher interpersonal uncertainty about one of the priors. Then, we derive testable predictions for the DM's decisions that maximize the expectation of a risk-averse U based on the DM's priors.

2.1 Assumptions about interpersonal uncertainty

For simplicity, we assume that the DM is probabilistically sophisticated and believes that dollars are valued non-negatively. Interpersonal uncertainty then means that the DM believes the per-dollar valuation of recipient j is distributed as $v_j \sim f_j$, where f_j is a probability distribution with non-negative support contained in $[0, b]$, and F_j is the corresponding CDF. For $x \geq 0$, we define $S_j(x) = \int_0^x F_j(y)$.

While we do not take a stance on the precise source of this uncertainty, we outline two plausible mechanisms that justify modeling v_j as a random variable from the DM's perspective. First, recipients may differ in their background wealth in ways unknown to the DM. Such *wealth heterogeneity* can affect how much recipients need or value additional money, even if they share the same utility function. Second, even among recipients with similar wealth, unobservable individual characteristics—such as intrinsic preferences⁶ or current needs—can cause *preference heterogeneity* and variation in the marginal utility of money. Such unobservable sources of heterogeneity might motivate the DM's subjective uncertainty over v_j , captured by the belief distribution f_j .⁷

Next, we assume that these belief distributions have two key features. First, a DM understands that different recipients might derive different values from the same allocated dollar amount based on their personalities, past experiences, socioeconomic status, or tastes. Thus, distributions over the valuations of others are non-degenerate.

Second, the belief distributions for different recipients systematically differ, depending on the DM's familiarity with the recipients, or the source of the \$100 endowment. For instance, suppose one recipient shares their hobbies, religious beliefs, or political interests with the decision-maker (ingroup member) while the other does not (outgroup member).

⁵Symmetry implies that the DM has no bias towards either recipient, which serves as our benchmark.

⁶For example, two equally wealthy recipients might have different utility curvatures.

⁷As detailed in Section 3.2, we directly measure interpersonal uncertainty and find robust evidence that DMs indeed perceive such uncertainty.

This makes the allocation decision an ingroup versus outgroup tradeoff, which will be our leading example. Facing this tradeoff, a DM might think that shared interests or identity with a recipient is indicative of shared past experiences, economic status, and tastes. As a consequence, DMs may be more confident or certain in assessing the individual characteristics of their ingroup, thus perceiving lower interpersonal uncertainty about the ingroup relative to the outgroup. Similarly, in situations where DMs themselves are one of the recipients, they may not perfectly know their own preferences⁸, but can assess them more confidently than they can assess others' preferences, leading to higher uncertainty about others' preferences. In other situations, DMs might think that one recipient is systematically more likely to have higher valuations than another recipient. Formally:

Definition 1. *DMs perceive higher interpersonal uncertainty for recipient 2 than recipient 1 if $S_1(x) \leq S_2(x)$ for all x with strict inequality, $S_1(y) < S_2(y)$, for some y . DMs perceive a mean-shifted interpersonal uncertainty for recipient 2a compared to recipient 2b if there exists $c \in \mathbb{R}_{++}$ such that for all x , $F_{2a}(x + c) = F_{2b}(x)$.*

The condition for higher interpersonal uncertainty is best understood as a generalization of “ f_2 is a mean-preserving spread of f_1 ” or equivalently “ f_2 is second-order stochastically dominated by f_1 ”, because the two quoted notions are defined identically with the additional condition that f_1 and f_2 have equal means. We use the concept of higher interpersonal uncertainty to characterize the optimal allocation x^* . In comparison, we use the mean-shift concept to study how x^* changes when the DM's beliefs about a particular recipient's valuation distribution shift to the right.

2.2 Choice behavior under interpersonal uncertainty

We investigate a benchmark case of unbiased utilitarian preferences, which means that the utility the DM receives from allocating $x \in [0, 100]$ to the ingroup and $(100 - x)$ to the outgroup is $v_1x + v_2(100 - x)$. As v_1, v_2 are random variables, she maximizes expected utility over the potential utilitarian outcomes:

$$EU(x) = E_{v_1 \sim f_1, v_2 \sim f_2} U(v_1x + v_2(100 - x)) \quad (1)$$

where $U' > 0$ and $E_{v_i \sim f_i}$ denotes the expectation with respect to f_i .

Given this setup, the optimal allocation depends crucially on the response to uncertainty as characterized by U , and on the belief distributions f_1 and f_2 . We will generally assume that $U'' < 0$ which implies that the DM dislikes higher variance over potential utilitarian outcomes.

⁸See Kreps (1979), Chakraborty (2021), and Gabaix and Laibson (2022) for models where DMs are uncertain about their own preferences.

Theorem 1. Suppose DM i has unbiased utilitarian preferences and is risk-averse ($U'' < 0$). If f_1 and f_2 are non-degenerate⁹, independent probability distributions, then

- i) Equal division: If $v_1 \stackrel{d}{=} v_2$ (i.e., $f_1 = f_2$), then i 's unique optimal allocation is $x^* = 50$.
- ii) Ingroup favoritism: If f_2 is a mean-preserving spread of f_1 , then i 's optimal allocation is $x^* \in (50, 100)$.
- iii) Comparative statics over x^* : Suppose the valuations of the two groups are distributed as v_1 and $c + v_2$ for some constant c and independent random variables $v_1 \sim f_1, v_2 \sim f_2$. Under arbitrary CARA preferences¹⁰, or under CRRA coefficient < 1 , the optimal allocation satisfies $dx^*/dc \leq 0$.

For the proof, see the Appendix. Part (i) follows from symmetry: a risk-averse DM hedges against interpersonal uncertainty by allocating equally among ex-ante symmetric recipients. (ii) shows that ceteris paribus, if the DM perceives higher interpersonal uncertainty about one of the recipients, she allocates more to the other recipient. Accordingly, a DM who perceives higher interpersonal uncertainty about the outgroup member will allocate more money to the ingroup, even if they believe that on average, ingroup and outgroup members benefit equally from receiving money. Similarly, a DM who perceives higher interpersonal uncertainty about others' utility than their own utility will allocate more money to themselves, even if they care about others equally and think that on average, everyone benefits equally from money. Finally, part (iii) shows that the DM would decrease the allocation to the ingroup (or the allocation to herself in the dictator game) if her belief about the outgroup's valuation mean-shifts to the right.¹¹

3 How interpersonal uncertainty shapes ingroup versus outgroup behavior

We empirically study how interpersonal uncertainty influences prosocial behavior in ingroup versus outgroup decisions. Our proposed mechanism has two parts: first, subjects are averse to interpersonal uncertainty; second, they perceive greater uncertainty for the outgroup than the ingroup. We propose that these two channels, combined, reinforce ingroup favoritism.

⁹If both f_1 and f_2 are degenerate, the DM will allocate 100 to the recipient with the higher value. If the degenerate distributions are equal, the DM is indifferent between all possible allocations.

¹⁰For a utility function $U(w)$, the coefficient of absolute risk aversion (ARA) is defined as $r_1(w) = \frac{-U''}{U'}$ and relative risk aversion (RRA) is defined as $r_2(w) = \frac{-wU''}{U'}$. CARA and CRRA imply r_1 and r_2 are constant, respectively.

¹¹Under extreme risk aversion, when c increases, the marginal return from the states with high v_2 is so low that on the margin, DMs might prefer to allocate more to v_1 to safeguard their utility in the states where v_2 is low.

Outline. This section proceeds in four steps. First, we isolate subjects’ *attitudes toward interpersonal uncertainty* by objectively varying levels of interpersonal uncertainty through synthetic groups. Second, we ask subjects directly about their *perceived interpersonal uncertainty* about outgroup and ingroup members, demonstrating a higher perceived uncertainty for outgroup members. Third, we combine these two components by calibrating objective uncertainty to match each subject’s own reported uncertainty perceptions. This allows us to quantify how much observed ingroup favoritism can be explained by interpersonal uncertainty and subjects’ response to it, separately from ingroup preferences. Fourth, we study how the joint effect of perceived uncertainty and attitudes toward it shapes behavior in a setting where ingroup preferences are absent by design.

Data. All experiments were preregistered and conducted on Prolific with participants living in the US between January 2024 and March 2026. See Appendix Table A.1 for an overview of all experiments conducted, their treatments, sample sizes, and purpose. See Appendix F for details on the mapping between preregistration and paper. We used oTree (Chen, Schonger, and Wickens, 2016) for programming the graphical user interface. Subjects spent a median of 10 to 15 minutes in the experiments and received between \$2.00 and \$2.50 as compensation (equivalent to an hourly wage between \$10 and \$15). Unless otherwise noted, one out of every ten subjects had one of their decisions implemented with real consequences (between-subject random incentivized system).¹²

3.1 Measuring aversion to interpersonal uncertainty

3.1.1 Experimental design

Paradigm. We use “bystander” money-allocation decisions to study ingroup versus outgroup decisions - one of the standard decision tasks used in the literature (e.g., Chen and Li, 2009; Enke, Rodríguez-Padilla, and Zimmermann, 2022).¹³ 120 subjects in our *Synthetic group* experiment participate as decision-makers (DMs), facing multiple such allocation decisions. In each decision, they are asked to allocate a fixed money amount between two groups of recipients. Between decisions, we vary (i) interpersonal uncertainty by varying the distribution of recipients’ valuation for money and (ii) ingroup preferences by varying recipients’ ingroup/outgroup affiliation. This requires us to measure ingroup/outgroup affiliations and valuation of money, for all recipients, which we explain next.

Measuring ingroup/outgroup affiliation. At the beginning of the experiment, all participants completed a short questionnaire in which they selected their top three hobbies from a

¹²In a meta-analysis, Umer (2023) shows that in the context of dictator games paying a subset of subjects instead of all subjects does not significantly change behavior.

¹³Past research has shown that behavior in bystander allocation decisions shows a high test-retest correlation, works equally well when posed hypothetically and incentivized, and is highly correlated with related psychological questionnaires (Enke, Rodríguez-Padilla, and Zimmermann, 2022).

provided list and reported their position on the political spectrum. Those assigned to the role of DM later played the allocation game and, in each decision, were matched to one ingroup recipient and one outgroup recipient. Depending on the decision, ingroup membership was defined either by shared hobbies/interests or by shared political views. DMs were therefore told that one recipient “shares your interests/hobbies” or “shares your political views (e.g., a fellow left-winger, or a fellow right-winger, etc.)”, while the other recipient “has different interests/hobbies than you” or “has different political views than you”.¹⁴

Measuring valuation for money. To exogenously vary interpersonal uncertainty, we provide DMs with a proxy for how much utility recipients receive from additional money. We measure this using recipients’ willingness-to-work (WTW) for a \$10 bonus payment. To elicit WTW, recipients faced a series of binary choices in a multiple-price-list format between (i) completing a required number of real-effort tasks to receive the \$10 bonus and (ii) opting out. Each real-effort task required moving 30 sliders to the middle position within a 60-second time limit, following Gill and Prowse (2012, 2019). A recipient’s WTW was defined as the number of tasks at which they first switched from preferring to complete the tasks to opting out. The idea is that, all else equal, individuals with a higher marginal utility of money should be willing to complete a higher number of these tedious and time-consuming slider tasks to obtain the bonus.

How subjects interpret WTW. To validate that subjects interpret WTW differences as informative about recipients’ valuation of money, we asked a separate set of subjects an open-ended follow-up question about how they interpret WTW differences.¹⁵ The modal response attributed WTW differences to wealth heterogeneity: 45.5% of subjects explicitly state that some recipients have a higher WTW because they need the money more. In addition, 37.2% mention reasons related to participants’ heterogeneous valuations of money. Overall, between 71.1% and 84.3% of responses are consistent with the interpretation of WTW differences as formulated in our framework. See Appendix Section C.1 for details.

Decisions. In each decision, DMs were endowed with \$10 to allocate between Group A and Group B. Each group consisted of two recipients who had participated in an earlier study in which their willingness-to-work (WTW) had been elicited.¹⁶ DMs were informed that, after their allocation choice, one recipient from each group would be randomly selected to receive the amount allocated to that group. Because only one recipient in each group

¹⁴We use the wording of Enke, Rodríguez-Padilla, and Zimmermann (2022).

¹⁵The question was asked at the end of the *Ingroup social/non-social* experiment using the effort environment, which also featured WTW differences.

¹⁶Recipients knew there was a 50% chance they would complete their selected WTW tasks and a 50% chance they would not work but might later receive money from DMs who observed their WTW. Only the latter group was matched with DMs in the main experiment. Thus, none had actually worked on the basis of their WTW choices, a fact clearly communicated to DMs. This ensured that fairness concerns based on realized effort did not affect the DM’s decisions.

was paid, variation in WTWs within a group creates uncertainty about the valuation of the person who ultimately receives the money.

Across decisions, we vary two features: whether recipients are ingroup or outgroup members, and how much WTW variation exists within each group. This yields four decision types:

1. *Uncertainty without group information* decision: the two recipients in Group A both have a WTW of 12, while the two recipients in Group B have WTWs of 4 and 22. Thus, the eventual recipient from Group A must have a WTW of 12, whereas the WTW of the eventual recipient from Group B is uncertain: it could be 4 or 22. No group identity information is provided.
2. *Group information without uncertainty* decision: the two recipients in Group A are ingroup members, while the two recipients in Group B are outgroup members. All four recipients have the same WTW of 12. Thus, group identity differs, but WTW variation is absent.
3. *High uncertainty on ingroup members* decision: the two recipients in Group A are ingroup members with WTWs of 4 and 22, while the two recipients in Group B are outgroup members who both have a WTW of 12. Thus, the ingroup is the more uncertain group.
4. *High uncertainty on outgroup members* decision: the two recipients in Group A are ingroup members who both have a WTW of 12, while the two recipients in Group B are outgroup members with WTWs of 4 and 22. Thus, the outgroup is the more uncertain group.

All decisions except the first were included twice: once when in/out-group affiliation was based on hobbies/interests and once on political views, yielding seven decisions in total.

Since WTW serves as a proxy for marginal utility, a higher spread in WTWs implies greater uncertainty about the marginal utility of the recipient who will receive the money, and hence greater interpersonal uncertainty. The *Uncertainty without group information* decision therefore identifies subjects' attitudes toward interpersonal uncertainty in the absence of ingroup preferences.¹⁷ Because the expected WTW in Group B exceeds that in Group A, namely $(22 + 4)/2 = 13 > 12$,¹⁸ a DM who nevertheless allocates more to Group A reveals a strict dislike of interpersonal uncertainty.

By contrast, the *Group information without uncertainty* decision identifies ingroup favoritism when interpersonal uncertainty is held constant. The final two decision types then reveal how shifting interpersonal uncertainty onto the ingroup or the outgroup changes the extent of ingroup favoritism.¹⁹

¹⁷DMs were familiar with the WTW concept because they first completed an example slider task and then underwent the WTW elicitation themselves.

¹⁸This argument holds both if recipients' valuation of the \$10 is directly equal to WTW and if valuation equals an increasing and convex transformation of WTW disutility. The latter assumption is well supported in the real-effort literature (see e.g., Gill and Prowse, 2019).

¹⁹When presenting WTW and social group information, we randomized each group's position on the screen

3.1.2 Results on allocation decisions

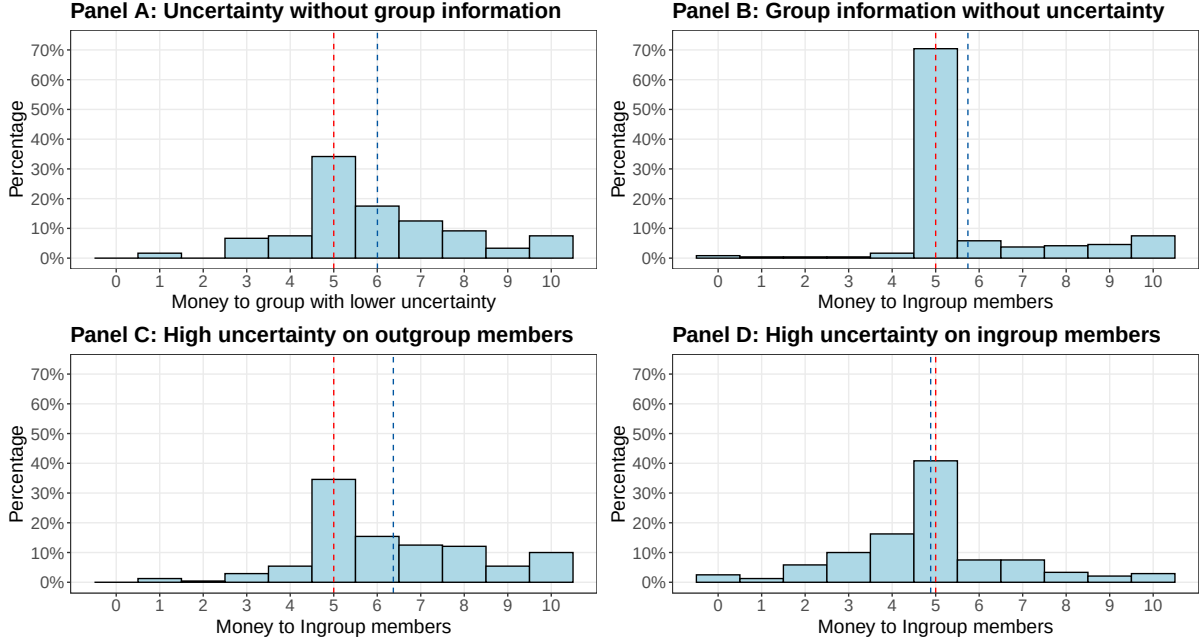
Distributions. Figure 2 displays the distribution of choices. Starting with Panel A, we observe that the majority of subjects are risk-averse towards interpersonal uncertainty even in the absence of group information: 54% allocate more than 50:50 to the group with lower WTW variance, 27% allocate 50:50 and a minority of 19% allocate more to the group with the higher WTW variance. On average, subjects allocate \$6.00 to the group with lower WTW variance, which is significantly more than the 50:50 benchmark ($p < 0.001$, one-sample Wilcoxon tests). In the case where all recipients have the same WTW (Panel B), subjects allocate on average \$5.74 to the ingroup (pooling the decision across both social groups), which is significantly different from the 50:50 split ($p < 0.01$, one-sample Wilcoxon test). This quantifies the extent to which ingroup preferences drive ingroup favoritism since interpersonal uncertainty about WTWs is absent in this decision.

Panels C to D then document how higher interpersonal uncertainty in the ingroup or the outgroup changes the magnitude of ingroup favoritism. The amount allocated to the ingroup increases from \$5.74 to \$6.37 when the outgroup is more uncertain (has higher variation in WTWs), and decreases to \$4.89 when the ingroup is more uncertain (both $p < 0.001$ compared to the no uncertainty decision, paired Wilcoxon tests). Moreover, changing the uncertainty changes the entire distribution of choices. When the outgroup has the higher WTW variance, the modal choice (62%) displays ingroup favoritism, while 12% of choices favor the outgroup and 26% are the 50:50 split. Once the recipients of both groups have equal WTWs, the modal choice (66%) is the 50:50 split. Here, the ingroup and outgroup are favored in only a minority of 29% and 5% of choices, respectively. Lastly, when the ingroup has the higher WTW variance, the modal choice (40%) switches to outgroup favoritism, while 31% of choices show ingroup favoritism and 29% are 50:50 choices. These results establish that responses to interpersonal uncertainty causally influence the extent of ingroup favoritism.

Regression analysis and structural model. To estimate the effects of uncertainty and preferences on ingroup favoritism, we regress indicator variables denoting the respective decision types on allocations made to the ingroup. We find that both ingroup preferences and interpersonal uncertainty significantly influence behavior. Pooled across groups, the coefficient on preferences is 0.666, and the coefficient on uncertainty is 0.742, both significant at the 0.001 level (see Appendix Table A.2). Accordingly, we find evidence for both influences. In Appendix Section C.2, we further estimate a structural model that jointly estimates subjects' risk aversion and ingroup preference. We estimate a CRRA parameter of $\gamma = 0.374$, which implies a slightly higher risk aversion toward interpersonal uncertainty than toward monetary risk.

(left or right) and the order in which the two pieces of information were shown.

Figure 2: The causal effect of interpersonal uncertainty on allocation decisions



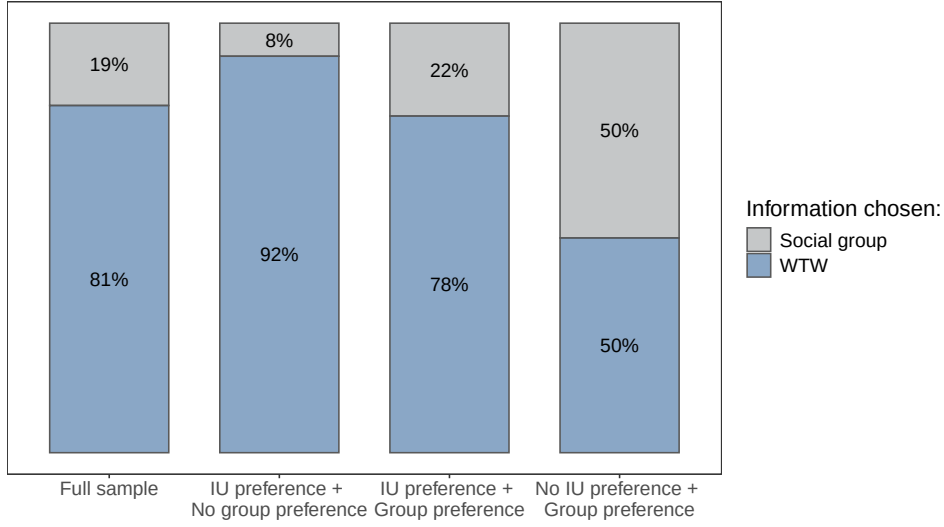
Notes: **Panel A:** The x-axis denotes the amount of money (out of \$10) allocated to the group with a lower variation in willingness-to-work (WTW). In **Panels B to D**, the x-axis denotes the amount allocated to the group whose members share a social group with the decision-maker instead of the group whose members are from a different social group. In Panel B, all group members have the same WTW. In Panel C, the outgroup has the higher WTW variation. In Panel D, the ingroup has the higher variation. The red line denotes the even split, the blue line the average allocation. In all panels, the binwidth is 1. Decisions are pooled across two social groups (shared hobbies/interests and political views), displaying $n = 120$ (Panel A) and $n = 240$ (Panels B to D) decisions made by 120 subjects.

Subject-level type analysis. We further exploit the within-subject structure to identify distinct behavioral types. We say a subject reveals a *group-based preference* if they choose a different allocation than 50:50 in *at least one* of the two decisions that provided *group information without uncertainty*. We say that a subject *responds to interpersonal uncertainty* if, for both social groups (interests and political views), their *outgroup uncertainty* decision was different from their *ingroup uncertainty* decision.²⁰ With this categorization, we find that 20% of subjects neither respond to uncertainty nor display a group preference. 33% of subjects respond to uncertainty but do not display a group preference, while 17% do not respond to uncertainty but display a group preference. Finally, 31% both respond to uncertainty and display a preference. Hence, interpersonal uncertainty is relevant for 64% of all subjects, while group preferences are relevant for 48%.

WTW-based similarity. Do DMs base their allocation decisions on how similar or dissimilar the recipients' WTWs are to their own, favoring recipients with more similar WTWs? For instance, a DM with a WTW close to 12 might prefer Group A (whose WTW is also 12)

²⁰Thus, we use a more conservative identification criterion for the response to uncertainty, because we require subjects to respond to uncertainty across both pairs of choices. In contrast, for identifying group-based preferences only one choice needs to be different from the “no favoritism” benchmark. In total, 79% of subjects respond to at least one change in uncertainty from ingroup uncertainty to outgroup uncertainty.

Figure 3: Demand for information on interpersonal uncertainty and group membership



Notes: Stacked bar charts showing the fraction of subjects who choose to receive the information about recipients' WTW, versus the information about recipients' group affiliation, before making an allocation decision. *Full sample* displays the information choice of all 120 subjects of the experiment. *IU preference + No group preference* displays the information choices among subjects who respond to interpersonal uncertainty but do not reveal an ingroup preference in their prior allocation decisions ($n = 39$). *IU preference + Group preference* displays the information choices among subjects who both respond to interpersonal uncertainty and reveal an ingroup preference ($n = 37$). *No IU preference + Group preference* displays the information choices among subjects who do not respond to interpersonal uncertainty but reveal an ingroup preference ($n = 20$).

simply due to this similarity. To test this, we define *WTW-distance* (i.e., dissimilarity) as the absolute difference between the DM's own WTW and Group A's WTW (12). Regressing the amount given to Group A on this *WTW-distance* reveals a weak and statistically insignificant relationship ($\beta = 0.017$, $p = 0.50$). Using alternative definitions of *WTW-distance* based on Group B WTWs (4 or 22) also yields non-significant results ($\beta = 0.02$, $p = 0.19$ and $\beta = -0.05$, $p = 0.16$, respectively). We therefore do not find evidence for WTW-similarity or dissimilarity driving the allocation choices.

Result 1. *Subjects are averse to interpersonal uncertainty, and this aversion shapes social decisions alongside social preferences.*

3.1.3 Results on information demand

A secondary prediction of interpersonal-uncertainty aversion is that subjects should value opportunities to resolve it. We study this through an information-choice task implemented immediately before the eighth and final allocation decision, following the seven allocation decisions described above.

Decision. In the final allocation decision, DMs knew that the two groups differed along two dimensions. First, one group consisted of ingroup members and the other of outgroup members. Second, one group had no WTW variation, with both recipients having a WTW of 12, whereas the other group had WTW values of 4 and 22. However, DMs did not know how

these two dimensions were assigned to the groups. Before making their allocation decision, they chose one dimension to reveal: either which group contained ingroup versus outgroup members, or which group had WTW variation.²¹

Results. As shown in Figure 3, in the full sample, 81% of subjects choose to learn the WTW information, whereas only 19% choose to learn the ingroup-outgroup information. Ingroup-outgroup affiliation is commonly viewed as a powerful indicator of social distance and a key determinant of economic and social behavior. However most subjects prefer to learn the WTW information instead.

Information demand also varies systematically across the behavioral types identified from subjects' earlier allocation decisions. Among subjects who respond to uncertainty but do not display a group preference, 92% choose the WTW information. This fraction falls to 78% among subjects who both respond to uncertainty and reveal a group preference, and falls further to 50% among subjects who reveal a group preference but do not respond to uncertainty. Thus, information demand aligns closely with revealed choice patterns: subjects whose prior decisions indicate a stronger role for interpersonal uncertainty are also more likely to actively seek uncertainty-relevant information. Taken together, these results support the interpretation that interpersonal uncertainty is a meaningful and often primary input into allocation decisions, rather than merely a form of motivated reasoning induced by group identity.

Result 2. *Information relevant for interpersonal uncertainty is not only attended to when supplied by the experimenter; it is actively demanded by subjects.*

3.2 Measuring perceived interpersonal uncertainty

The synthetic-group paradigm shows that subjects are averse to objectively induced interpersonal uncertainty. The second part of our mechanism concerns beliefs: do subjects perceive more interpersonal uncertainty about outgroup members than ingroup members? To answer this question, we directly elicit perceived interpersonal uncertainty.

Elicitation. We elicited interpersonal uncertainty in several experiments, both to relate perceived uncertainty to our main behavioral outcomes and to assess the robustness of the measure across contexts. We start with the elicitation from the *Synthetic group calibration* experiment (120 subjects). This elicitation provides the cleanest setting for the measure because subjects first received detailed instructions explaining certainty and uncertainty pertaining to the elicitation task, and then were tested on their comprehension.

On the belief elicitation screen, subjects were presented with descriptions of an ingroup and an outgroup, using the same wording and social groups as in the previous section. They

²¹We balanced the presentation of both pieces of information by randomizing the order in which the information was introduced and displayed for the choice.

were then instructed to form a prediction in their mind about how strongly an individual from each group would value receiving \$10. We then elicited interpersonal uncertainty separately for ingroup and outgroup with the following question (using interests/hobbies as illustration):

“Consider individuals who [share your interests/hobbies] OR [have different interests/hobbies than you]. How certain do you feel about your prediction of how strongly this person would value receiving \$10?

Subjects responded on an 11-point scale where 0 and 10 were labeled as *Very uncertain* and *Very certain*, respectively. We recode the variable so that higher values indicate greater uncertainty. We then construct the following individual-level measure of *relative uncertainty*:

$$\text{Relative uncertainty} = \text{Uncertainty about outgroup valuation} \\ - \text{Uncertainty about ingroup valuation}$$

Positive values therefore indicate higher uncertainty about outgroup valuations.

Results. We find that a large majority of subjects perceive non-zero uncertainty about outgroup members’ valuations (87% for hobbies/interests, 88% for political views) and ingroup members’ valuations (79% for hobbies/interests, 80% for political views). Importantly, this uncertainty is significantly higher for the outgroup than for the ingroup ($p < 0.001$, paired Wilcoxon tests). As displayed in the first two rows of Table 1, subjects perceive uncertainty that is 1.81 (1.71) scale points higher for outgroup members in the case of hobbies/interests (political views). With respect to the distribution of differences at the individual level, in 53% (52%) of cases, subjects report higher uncertainty for the outgroup; in 3% (5%) they report higher uncertainty for the ingroup, and in the remaining 44% (43%) of cases, subjects report no difference. These results provide evidence for the second part of our mechanism.

Robustness. We elicited perceived uncertainty in the following additional experiments, varying design features to assess robustness.

- In *Ingroup effort social* (121 subjects), we elicited uncertainty without defining uncertainty jointly with comprehension questions. Instead, subjects first made allocation decisions between an ingroup member and an outgroup member (see Section 3.4.1).²²
- In *Ingroup social* (119 subjects), the setup was the same as in *Ingroup effort social*, except that the reward medium differed. Instead of a bonus payment, the question asked subjects about their certainty about others’ valuation for Amazon gift card money.

²²The question was also worded slightly differently: “How certain are you about how individuals who [share your interests/hobbies] OR [have different interests/hobbies than you] would value the bonus payment?”

Table 1: Interpersonal uncertainty across social groups and paradigms

Social group	Part of treatment	Outgroup IU - Ingroup IU			
		Average diff.	% > 0	% = 0	% < 0
Hobbies/interests	<i>Synthetic group calibration</i>	1.81	0.53	0.44	0.03
Political views	<i>Synthetic group calibration</i>	1.71	0.52	0.43	0.05
Robustness					
Hobbies/interests	<i>Ingroup IU belief</i>	0.81	0.42	0.51	0.07
Hobbies/interests	<i>Ingroup social</i>	0.57	0.34	0.55	0.11
Hobbies/interests	<i>Ingroup effort social</i>	0.95	0.34	0.62	0.04
Political views	<i>Ingroup IU belief</i>	0.93	0.36	0.59	0.05
Political views	<i>Ingroup social (gift card)</i>	1.01	0.33	0.54	0.13
Political views	<i>Ingroup social (real effort)</i>	1.39	0.40	0.53	0.07
Religious beliefs	<i>Ingroup IU belief</i>	0.66	0.34	0.63	0.03
Religious beliefs	<i>Ingroup social (gift card)</i>	0.45	0.25	0.64	0.11
Religious beliefs	<i>Ingroup social (real effort)</i>	1.01	0.27	0.68	0.05

Notes: The table reports interpersonal uncertainty differences by social group and paradigm. Difference in interpersonal uncertainty is defined as reported uncertainty about outgroup valuation minus reported uncertainty about ingroup valuation. "Average diff." denotes the mean uncertainty difference. The columns > 0, = 0, and < 0 report the shares of subjects with positive, zero, and negative uncertainty differences, respectively.

- Lastly, *Ingroup IU belief* (60 subjects) was the same as *Ingroup social* except for two differences: first, it featured no prior allocation choice. Instead, prior to eliciting uncertainty, we elicited subjects' beliefs about the average valuation of \$100 in Amazon gift card money of ingroup and outgroup members. Specifically, subjects were asked: "Consider individuals who [share your interests/hobbies] OR [have different interests/hobbies than you]. On average, how much do you think they value [their bonus reward]?"

Robustness results. Table 1 reports the results of the additional experiments. Across all paradigms and social groups, average relative uncertainty is positive. The magnitude varies across experiments, but the sign is stable: subjects report greater uncertainty about outgroup valuations than about ingroup valuations. Similarly, the share of subjects reporting higher uncertainty about the outgroup is substantially larger than the share of subjects showing the reverse pattern across experiments.

Beliefs about average valuations. Using *Ingroup IU belief*, we can compare this asymmetry in uncertainty about valuations with beliefs about average valuations. Here, subjects report ingroup members to have valuations of \$87.85 (hobbies/interests), \$89.64 (political views), and \$89.14 (religious beliefs). For outgroup members, the corresponding reported valuations are \$88.20, \$86.69, and \$87.85. Thus, on average, subjects' beliefs do not differ between ingroup and outgroup members ($p = 0.90$, $p = 0.06$, and $p = 0.37$, paired

Wilcoxon tests), and are well calibrated, as the actual average valuation observed in the experiment is \$87.

Validation exercise. As an additional validation exercise, at the end of the *Synthetic group paradigm* experiment (120 subjects), we presented subjects with two artificial groups featuring exogenously induced WTP distributions, one of which was a mean-preserving spread of the other. Subjects then reported their perceived interpersonal uncertainty for each group. Overall, 74% (87%, respectively) of subjects report strictly (weakly) higher interpersonal uncertainty, validating the sensitivity of the uncertainty measure. See Appendix D.1 for details.

Result 3. *Subjects report systematically higher interpersonal uncertainty about outgroup than ingroup members across elicitation contexts. This difference appears in uncertainty about valuations but not in beliefs about average valuations.*

3.3 Measuring the importance of interpersonal uncertainty

The previous two sections separately identified the two parts of our mechanism: subjects' attitudes toward interpersonal uncertainty, and their perceived uncertainty about ingroup versus outgroup members. The next question is quantitative: how much of observed ingroup favoritism can these two components explain?

Answering this requires linking the two measures. The synthetic-group design identifies subjects' attitudes toward objective WTW spreads, but the fixed spreads used in that design need not match the amount of uncertainty subjects perceive about ingroup and outgroup members. We therefore introduce a calibration design that lets us ask how subjects behave when the uncertainty difference they report between ingroup and outgroup members is recreated in synthetic groups. This provides a direct estimate of the magnitude of interpersonal uncertainty relative to ingroup preferences.

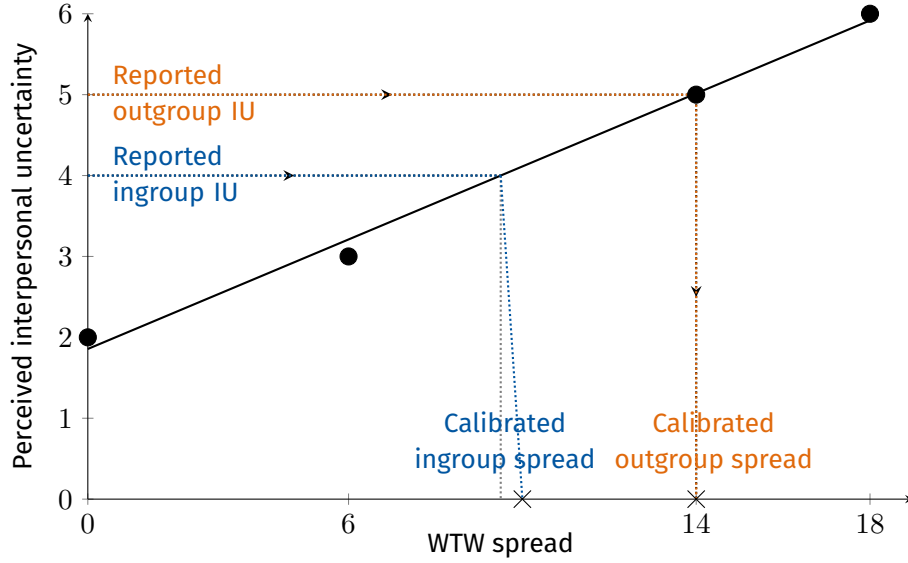
3.3.1 Experimental design

As before, subjects first report their social-group membership and familiarize themselves with the WTW measure by reporting their own WTW. The design then proceeds in three steps:

i) Perceived IU elicitation. As described in the previous section 3.2, subjects indicate their certainty about ingroup and outgroup members' valuation of \$10. The difference between these two ratings yields our individual-level measure of *relative interpersonal uncertainty*.

ii) Calibrating IU to objective WTW spreads. We next measure how each subject translates *objective* variations in WTW valuations into *perceived* uncertainty. Subjects are shown four synthetic groups, each consisting of two earlier participants whose WTW for earning

Figure 4: Illustration of the calibration procedure



Notes: The figure illustrates the calibration procedure for an example subject. The four black dots show the subject's reported interpersonal uncertainty for synthetic groups with WTW spreads of 0, 6, 14, and 18. The fitted line shows the subject-specific OLS relationship between objective WTW spread and perceived uncertainty, given by $\widehat{IU} = 1.86 + 0.23x$. The two horizontal dotted lines represent the subject's reported uncertainty for ingroup and outgroup members. Projecting these reports onto the fitted line yields the WTW spreads of the calibrated ingroup and outgroup synthetic groups. The projections are then approximated to the nearest integer. In this example, the ingroup and outgroup spreads are calibrated to 10 (from 9.50) and 14 (from 13.93), respectively.

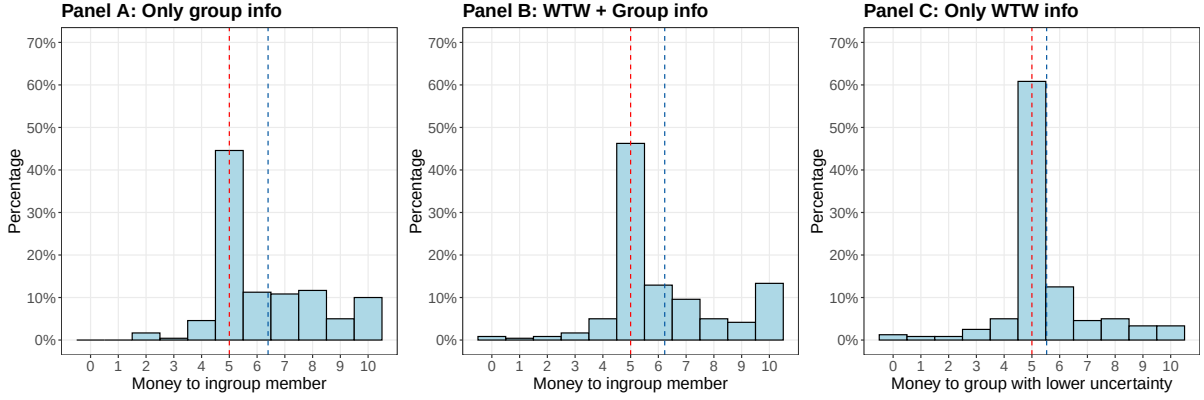
\$10 had been elicited. Across these groups, we vary the WTW spread as follows: (20, 20), (18, 24), (14, 28), and (12, 30).²³ For each synthetic group, subjects report how certain they are about the valuation of a randomly selected member, using exactly the same question wording and 11-point scale as in the belief elicitation.

Together, the first and second stages deliver the two inputs required for calibration. The first tells us how much more uncertainty a subject perceives about outgroup than ingroup members. The second tells us how strongly that subject's perceived uncertainty responds to objective WTW spreads. We estimate this mapping at the individual level using a linear regression of reported uncertainty on WTW spread. We then use the estimated mapping to construct, for each subject, synthetic groups whose WTW spreads reproduce the uncertainty levels the subject reported for the ingroup and the outgroup. We refer to these as *socially calibrated synthetic groups*. For an illustration of the calibration, see Figure 4. Appendix Section C.3 provides the technical details.

iii) Allocation decisions. In each decision, DMs allocate \$10 between socially calibrated synthetic groups A and B. We construct these groups separately for each ingroup-outgroup distinction. For each subject and each distinction, we construct one synthetic group whose WTW spread matches the subject's reported uncertainty about ingroup members, and an-

²³The calibration relies only on differences in WTW spreads, not differences in average WTW. The lower mean of the no-spread group is retained to maintain similarity with the original synthetic-group treatment.

Figure 5: Allocation decisions in calibrated interpersonal uncertainty choice tasks



Notes: **Panel A** and **Panel B:** The x-axis denotes the amount of money (out of \$10) allocated to the group whose members share a social group with the decision-maker instead of the group whose members are from a different social group. **Panel C:** The x-axis denotes the amount allocated to the group with a lower variation in willingness-to-work (WTW). In Panel A, decision-makers know only the social group affiliation of recipients. In Panel B, they know both group affiliation and the WTW of each group member, with the values calibrated to their perceived uncertainty. In Panel C, they know only the calibrated WTW of each group member. The red line denotes the even split, the blue line the average allocation. In all panels, the binwidth is 1. Decisions are pooled across two social groups (shared hobbies/interests and political views), displaying $n = 240$ decisions made by 120 subjects.

other synthetic group whose WTW spread matches the subject's reported uncertainty about outgroup members. For each distinction, DMs face three allocation decisions. We do this for hobbies/interests and for political orientation, resulting in six decisions in total, presented in random order.

1. *Only WTW info* decision: Subjects observe the calibrated WTW values of group members in each group but receive no information about ingroup or outgroup status.
2. *Only group info* decision: Subjects learn that the two recipients of Group A are ingroup members, while the two recipients of Group B are outgroup members. No information about WTW values is given.
3. *WTW + group info* decision: Subjects receive both pieces of information: the calibrated WTW values of group members and their ingroup/outgroup status.

As before, subjects are told that money allocated to a group goes to a randomly chosen person from the pair.

3.3.2 Results

Perceived uncertainty. As reported in Section 3.2, subjects perceive significantly higher interpersonal uncertainty about outgroup members than about ingroup members. Regarding the WTW spreads for the calibration, subjects report an average uncertainty of 1.18 for the (20, 20) spread, 2.97 for the (18, 24) spread, 4.79 for the (14, 28) spread, and 6.18 for the (12, 30) spread. Accordingly, the average calibrated WTW spread for the ingroup is 5.71, while it is 10.71 for the outgroup. For a distribution of the spreads, see Appendix

Figure A.1.²⁴

Calibration replicates ingroup favoritism. In both *Only group info* and *WTW + group info* decisions, we replicate ingroup favoritism, with giving significantly differing from the 50:50 benchmark (both: $p < 0.001$, one-sample Wilcoxon tests). If the calibration procedure successfully recreates perceived interpersonal uncertainty using calibrated synthetic groups, then we should see no strong differences between the two decisions. Consistent with this prediction, average giving to the ingroup is \$6.40 in *Only group info* and \$6.23 in *WTW + group info*, a non-significant difference ($p = 0.11$, one-sample Wilcoxon test). Likewise, the distributions of choices appear similar, as displayed in Panel A and Panel B of Figure 5.

Risk aversion toward interpersonal uncertainty. Next, we move to the *Only WTW info* decision, which measures risk attitudes toward the calibrated level of uncertainty. On average, subjects give \$5.53 to the group with the smaller WTW spread, which is significantly different from the 50:50 benchmark ($p < 0.001$, one-sample Wilcoxon tests). As Panel C of Figure 5 shows, in 33% of decisions subjects allocate more to the smaller-spread group, in 57% the 50:50 split is implemented, and in 10% of cases subjects give more to the higher-spread group.²⁵ Accordingly, we replicate aversion to interpersonal uncertainty when the objective uncertainty is calibrated to subjects' own perceived ingroup/outgroup uncertainty.

Relative magnitude. Comparing ingroup favoritism in *Only group info* and *Only WTW info* allows us to assess the magnitude of interpersonal uncertainty in driving ingroup favoritism. Specifically, the average allocation to the ingroup in *Only group info* equals \$6.40, which is a combination of both interpersonal uncertainty and ingroup preferences. In *Only WTW info*, where ingroup preferences cannot operate, subjects give \$5.53 to the group with the WTW spread calibrated to the outgroup. This implies that $0.53/1.40 = 38\%$ of ingroup favoritism can be explained by interpersonal uncertainty. Note that this is a conservative estimate: If some subjects do not view WTW as a proxy for valuation, they will perceive no uncertainty differences and therefore choose 50:50 allocations, attenuating the estimated effect.

Result 4. *Interpersonal uncertainty is quantitatively important for ingroup versus outgroup decisions: at least 38% of observed ingroup favoritism can be explained by interpersonal uncertainty.*

²⁴We can only estimate spreads for 120 of the 170 subjects, who report monotonically higher uncertainty as the spreads increase. Importantly, reported uncertainty about ingroup and outgroup members does not differ between 120 subjects for whom estimation is possible and the 50 for whom it is not, suggesting that this restriction does not systematically bias our quantification exercise. See Appendix C.3 for details.

²⁵The large share of equal splits partly reflects the calibration itself. In 48% of decisions, the two calibrated groups have the same WTW spread; in these cases, 87% of subjects choose the 50:50 split. When the calibrated spreads differ, behavior shifts toward the lower-uncertainty group: in 52% of decisions, subjects allocate more than \$5 to the smaller-spread group, 30% choose the 50:50 split, and 17% allocate less than \$5. Moreover, giving to the lower-uncertainty group is significantly correlated with ingroup favoritism in *Only group info* ($r = 0.31$, $p < 0.001$).

3.4 Combining beliefs and risk attitudes: the non-social paradigm

The non-social paradigm provides an independent check on our main findings. The synthetic-group and belief-measurement exercises require linking attitudes toward objective uncertainty to reported uncertainty about ingroup and outgroup members. The non-social paradigm does not rely on this calibration. Instead, it removes the social component of the allocation decision while preserving the relevant subjective uncertainty. We use this paradigm to test whether subjects display similar behavioral patterns in decisions that do not involve monetary consequences for ingroup or outgroup members relative to those that feature consequences.

3.4.1 Experimental design

The experimental sessions using the ingroup-outgroup paradigm feature two distinct decision situations, *social* and *non-social* decisions.

Ingroup social decisions. For the *Ingroup social* decision, we again use the “bystander” money-allocation game, similar to the experiment in Section 3.1.1. Each subject allocates $\$M$ between an ingroup and an outgroup member thrice, once for each social group (hobbies/interests, political views, and religious affiliation).

Ingroup non-social decisions. We design a *non-social* decision that removes consequences for other individuals while preserving uncertainty about their valuations. In this task, the DM’s choice affects only their own payoff. The DM allocates $\$M$ between two payoff components, one tied to an ingroup member’s valuation and one tied to an outgroup member’s valuation. The DM’s payoff is given by

$$\Pi(x_{in}, x_{out}) = x_{in} \cdot v_{in}/M + (M - x_{in}) \cdot v_{out}/M.$$

The values v_{in} and v_{out} denote the ingroup and outgroup members’ valuation for the $\$M$ reward, elicited by the researchers, but unknown to the DM. Dividing by M converts each valuation into a valuation per dollar of the reward.

Since DMs do not know the actual valuations of the matched individuals, such interpersonal uncertainty transforms the *non-social* decision into a subjective lottery choice. Importantly, the ingroup and outgroup members’ valuations enter the payoff function symmetrically, so any differences in x_{in} and $x_{out} = M - x_{in}$ are driven only by differences in beliefs and uncertainty about the valuations.

Eliciting valuations. For robustness, we implement the *social* and *non-social* decisions in two reward environments across sessions, *Effort* and *Gift card*. The environments differ in how v_{in} and v_{out} are measured.

In the *Effort* session, subjects allocated and split $\$M = 10$ in the *social* and *non-social* tasks, respectively. In the *social* tasks, the ingroup and outgroup members receive the allo-

cated money as bonus payment after the experiment. In the *non-social* tasks, the valuations v_{in} and v_{out} used to incentivize the DMs corresponded to the ingroup and outgroup recipients' respective willingness-to-work (WTW) for the \$10 bonus payment. We elicit and describe the WTW exactly as done in Section 3.1.1. As in the synthetic-group paradigm, WTW serves as a proxy for recipients' valuations of the bonus. Here, we use these valuations to determine DMs' payoffs in the *non-social* decisions. To understand how the DMs were incentivized using the WTWs, suppose the DM split \$8 and \$2 in favor of the ingroup and outgroup member, respectively. If the elicited WTW_{in} and WTW_{out} were 20 and 10, then the DM's payoff would be

$$\Pi(8, 2) = 8 \cdot 20/10 + 2 \cdot 10/10 = 18.$$

Instead, if the WTW_{in} and WTW_{out} were 10 and 20, respectively, the payoff would become 12. Since DMs do not know the exact WTW_{in} and WTW_{out} , they face uncertainty about the payoff consequences of allocating to the ingroup- versus outgroup valuations.

The *Non-social* decision of the *Gift card* environment has the same structure, with two differences: here, subjects allocate gift card money, and the valuations v_{in} and v_{out} correspond to the recipients' willingness-to-pay (WTP) for gift cards. Specifically, in the *social* tasks, DMs allocate $M = \$100$ that is sent to the ingroup and outgroup member six weeks from the date of the experiment in the form of Amazon gift card money. In the *non-social* tasks, the elicited WTPs measured subjects' present-day dollar valuation of the delayed gift card money using a standard incentivized multiple-price-list (MPL) procedure.²⁶

Minimizing inattention and confusion. We employ several measures to mitigate inattention and confusion. First, decision-makers complete the relevant WTW or WTP valuation task themselves before completing the *non-social* decisions. This familiarizes them with the valuation process that underpins the incentive. Second, we include several comprehension questions that test whether DMs understood that the *non-social* decisions only have consequences for themselves, not for the other individuals. If they answer incorrectly on the first attempt, they are informed of the mistake and asked to try again. If they still err, we explain the correct answers and clarify the key concepts. This process reinforces the distinction between *social* and *non-social* decisions. Third, to further minimize inattention, we include an explicit disclaimer on the *non-social* decision screens that states

“Reminder: your choice only determines your own payment, it does not affect the two individuals.”

On the decision screen, we also provided DMs with the option to revisit the instructions.

²⁶Subjects made a series of binary decisions between (i) receiving a \$100 Amazon gift card in six weeks and (ii) receiving a monetary amount today, which increased across decisions. We enforced single switching by automatically filling the list above and below subjects' choices. The six-week delay was introduced to further distinguish the valuation object - gift card money - from the medium used to elicit it - plain bonus money.

Procedure. We first ran an experimental session in the *Gift card* environment with 119 subjects, and then four months later a session using the *Effort* environment with 121 subjects. Within each session, we randomized the order of decisions. Half of the decision-makers first faced the *social* decision and then the valuation task and *non-social* decision. The other half first faced the valuation task and *non-social* decision, and subsequently the *social* decision. We did not announce beforehand that other decisions would follow the initial decisions, therefore minimizing the scope for contagion from one treatment to the other. This design allows us to analyze within-subject behavior and compare between-subject behavior by only looking at the first set of decisions. As before, in the *Effort* environment, one out of every ten subjects had one of their decisions implemented with real consequences; in the *Gift card* environment (and subsequent experiments that use WTPs), one subject out of all participating subjects had one of their decisions implemented.

3.4.2 Results

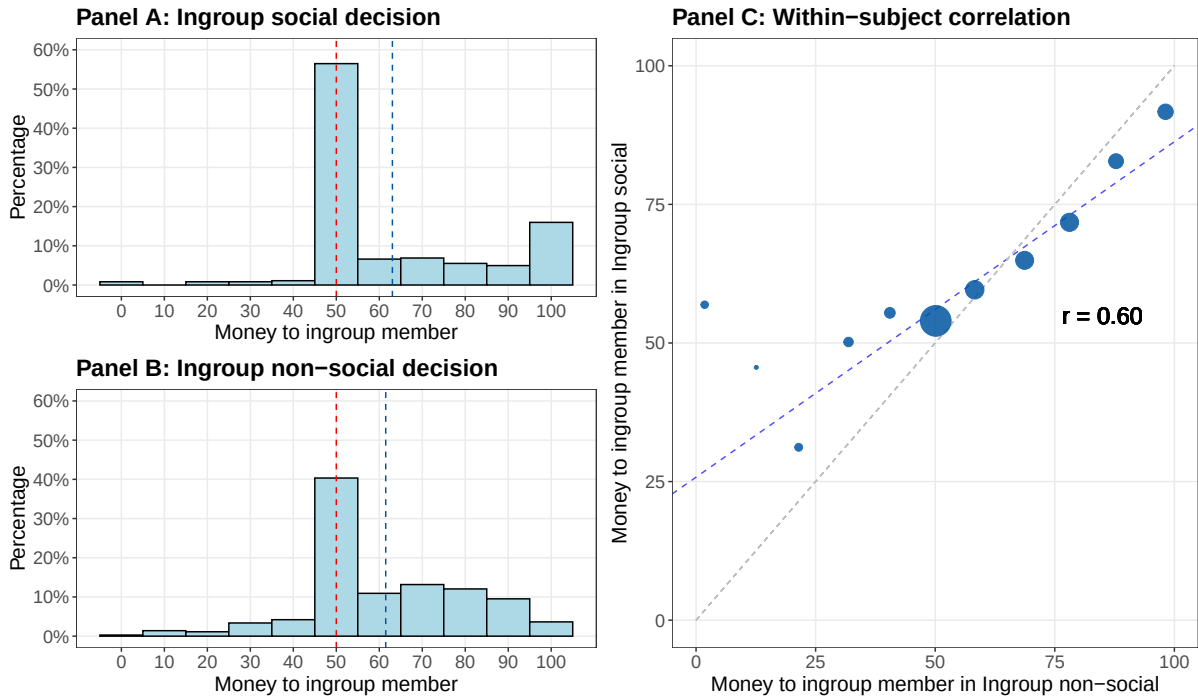
We pre-registered the *Gift card* and *Effort* environments separately. However, as the results are quantitatively almost identical, we present the combined results here and the separate results in Appendix E.1.²⁷ We utilize a between-subject comparison of the *social* and *non-social* decisions.

Ingroup social decisions. In the *Ingroup social* decisions, subjects allocate on average \$57.75, \$70.66, and \$60.73 to the ingroup member when they share the same interests/hobbies, the same political views, or the same religious beliefs, respectively. In all three cases, we can reject the hypothesis of no ingroup favoritism ($p < 0.001$, one-sample Wilcoxon tests). Panel A of Figure 6 displays the distribution pooled over the three decisions, which replicates the typical distributional pattern found in the literature (e.g., Enke, Rodríguez-Padilla, and Zimmermann, 2022). Subjects allocate strictly more than half to the ingroup in 44% of decisions, strictly more to the outgroup in 5%, and split evenly in 51%. Overall, 71% of subjects display ingroup favoritism in at least one decision.

Ingroup non-social decisions. A similar pattern appears in the *Ingroup non-social* decisions, even though these choices have no consequences for ingroup or outgroup members. Here, subjects allocate on average \$58.61, \$64.60, and \$61.29 in favor of the valuation of the ingroup member who shares their interests/hobbies, political views, and religious beliefs. All three averages are significantly above 50:50 ($p < 0.001$, one-sample Wilcoxon tests). Panel B of Figure 6 displays the pooled distribution. Subjects allocate strictly more than half to the ingroup-linked valuation in 59% of decisions, strictly more to the outgroup-linked valuation in 12%, and split evenly in 29%. Overall, 78% of subjects display ingroup-favoring behavior in at least one *non-social* decision.

²⁷We multiply the *Effort* decisions by 10 to ensure a common scale with the *Gift card* decisions.

Figure 6: Main results ingroup versus outgroup decisions



Notes: Panels A and B: Histograms of *Ingroup social* (Panel A) and *Ingroup non-social* (Panel B) decisions. The binwidth is 10. In Panel A, the x-axis denotes the amount of money (out of \$100) allocated to the ingroup member instead of the outgroup member, with decisions having consequences for the ingroup and outgroup members. In Panel B, the x-axis denotes the amount (out of \$100) allocated to the ingroup member's valuation instead of the outgroup member's valuation, where the decisions have consequences only for the decision-makers, with their payoff depending on the ingroup and outgroup members' valuations of money. The red line denotes the even split, the blue line the average allocation. **Panel C:** Binned scatter plot of *Ingroup social* and *Ingroup non-social* decisions. We divide *Ingroup non-social* choices into 10 bins and plot the average giving to the ingroup member in *Ingroup social* for each bin. The blue line displays the linear fit of regressing *Ingroup social* on *Ingroup non-social* decisions. The correlation coefficient is $r = 0.60$. Decisions are pooled across the three groups (shared hobbies/interests, political views, and religious beliefs), displaying $n = 363$ (Panel A), $n = 357$ (Panel B), and $n = 720$ (Panel C) decisions by 121, 119, and 240 subjects, respectively.

Comparing *Ingroup social* and *non-social*. The main difference between the two distributions is the frequency of maximal ingroup favoritism. In the *social* decisions, 15% of subjects allocate the entire endowment to the ingroup member, compared to only 3% in the *non-social* condition. The distributions are therefore significantly different ($p = 0.01$ for hobbies/interests, $p = 0.001$ for political views, $p = 0.01$ for religious beliefs; Kolmogorov-Smirnov tests). This pattern suggests that extreme ingroup favoritism in *Ingroup social* decisions reflects both preferences and beliefs, consistent with the evidence from the synthetic-group paradigm.

Within-subject comparison. We include the second set of subjects' decisions to perform the within-subject analysis – this data replicates the between-subject results (see Appendix E.2.1 for details) and shows no evidence of order effects (see Appendix E.3). Panel C of Figure 6 uses a binscatter to plot each individual's social decision against their non-social decision, and shows that ingroup-favoring behavior in *non-social* strongly predicts ingroup

favoritism in *social* ($r = 0.60, p < 0.001$).²⁸ Thus, the same subjects who favor the ingroup when choices affect others also favor the ingroup-linked valuation when choices affect only their own payoff.²⁹

Result 5. *Subjects display ingroup-favoring behavior even in Ingroup non-social decisions, which retain interpersonal uncertainty but remove any consequences for others. Moreover, this behavior strongly predicts ingroup favoritism in Ingroup social decisions.*

3.4.3 Potential confounds and robustness treatments

The correlation between *Ingroup social* and *Ingroup non-social* decisions could reflect mechanisms other than interpersonal uncertainty. We consider four such alternative explanations:

- *Confusion*: Subjects might find the *non-social* incentives confusing or too similar to *social* incentives, leading them to use same choice rule in both conditions.
- *Indifference*: Subjects might be indifferent between all *non-social* allocations, and use their *social* choices as a tiebreaker. This is only possible if the beliefs about ingroup and outgroup valuations are degenerate and exactly equal.
- *Loyalty*: They may use both tasks to signal loyalty to the ingroup.
- *Social Framing*: The residual social wording in the *non-social* might activate social considerations or heuristics.

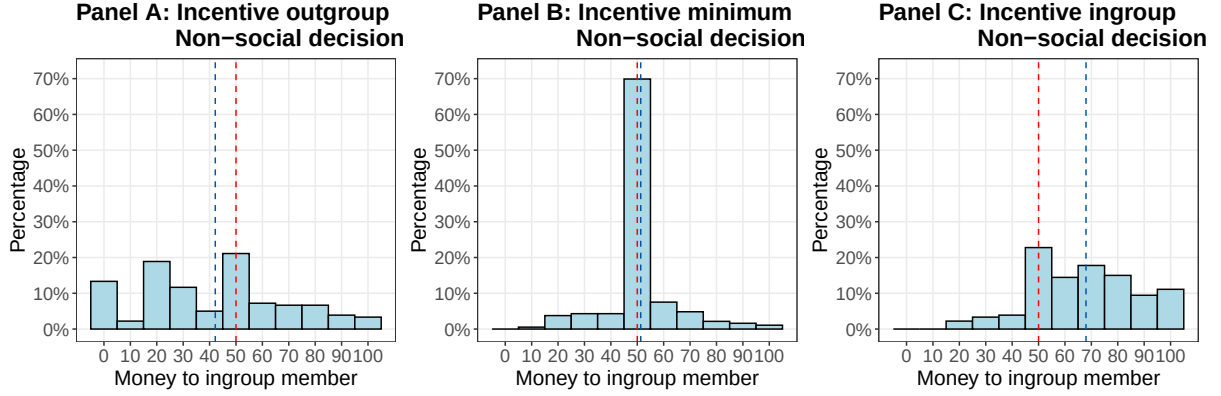
We address these concerns in two steps. First, we use comprehension quiz data to assess comprehension. Second, we implement four robustness treatments that vary incentives or remove the social framing altogether.

Comprehension-based evidence against confusion. We sort subjects by their performance on the comprehension quiz, which explicitly included questions on the *non-social* payoff structure. Pooling decisions across the three social groups, subjects with at most one error allocate \$62.32 in *Ingroup social* and \$60.90 in *Ingroup non-social* to the ingroup member. Subjects with more than one error allocate \$64.07 and \$62.50, respectively. Average allocations also remain significantly above 50:50 in within each social group, and under alternative comprehension-based sample splits; see Appendix Figure A.2.

²⁸The correlation is not driven by subjects who display no or maximal ingroup favoritism; excluding either group yields correlations of similar or larger magnitude.

²⁹As Panel A of Appendix Figure A.4 shows, perceived interpersonal uncertainty is also significantly correlated with *Ingroup non-social* decisions ($r = 0.23, p < 0.001$).

Figure 7: Incentive ingroup and Incentive outgroup robustness treatment results



Notes: Histogram of *Incentive outgroup* (Panel A), *Incentive minimum* (Panel B), and *Incentive ingroup* (Panel C) decisions, where subjects maximize $(x_{in} \cdot WTP_{in}/100 + 3 \cdot x_{out} \cdot WTP_{out}/100)$, $\min\{x_{in} \cdot WTP_{in}/100, x_{out} \cdot WTP_{out}/100\}$, and $(3 \cdot x_{in} \cdot WTP_{in}/100 + x_{out} \cdot WTP_{out}/100)$, respectively. The x-axis denotes the amount of gift card money (out of \$100) that subjects allocate to the ingroup member's WTP instead of the outgroup member's WTP. The red dotted line denotes the even split benchmark, the blue dotted line the average allocation. The binwidth is 10. Decisions are pooled across the three groups (shared hobbies/interests, political views, and religious beliefs). Panels A and C display $n = 180$ decisions by 60 subjects; Panel B displays $n = 186$ decisions by 62 subjects.

Robustness treatments addressing confusion, indifference, and signaling concerns.

We implement three variants of the *Ingroup non-social* condition that change only the payoff function. In *Incentive outgroup*, the outgroup-linked valuation receives triple weight:

$$\Pi(x_{in}, x_{out}) = x_{in} \cdot WTP_{in}/100 + 3 \cdot x_{out} \cdot WTP_{out}/100$$

Conversely, in the *Incentive ingroup* condition, the weight on the ingroup member is tripled:

$$\Pi(x_{in}, x_{out}) = 3 \cdot x_{in} \cdot WTP_{in}/100 + x_{out} \cdot WTP_{out}/100$$

In the third treatment – *Incentive minimum* – we replace the utilitarian objective with a Rawlsian one. Here, subjects' payoffs are calculated as:

$$\Pi(x_{in}, x_{out}) = \min\{x_{in} \cdot WTP_{in}/100, x_{out} \cdot WTP_{out}/100\}$$

Predictions. Under our framework – utilitarianism subject to interpersonal uncertainty – the predictions are straightforward: *Incentive outgroup* should increase allocations towards the outgroup-WTP. *Incentive ingroup* should increase allocations towards the ingroup-WTP. *Incentive minimum* should lead to 50 : 50 allocations.³⁰

In contrast, *confusion* and *loyalty* accounts predict consistent displays of ingroup-favoring behavior across all three treatments. *Indifference* predicts allocating the entire endowment of 100 to either the outgroup or ingroup WTP.

³⁰As shown in Appendix B, even if the outgroup WTP distribution is a mean-preserving spread of the ingroup WTP distribution and subjects are risk-averse, the predicted optimum under the Rawlsian objective and our assumptions is $(x_{in} = x_{out} = 50)$.

Results. Figure 7 presents the results. Relative to an average allocation of \$60.89 to the ingroup-WTP in *Ingroup non-social*³¹, allocations shift strongly in the directions predicted by our framework. Subjects allocate \$42.16 to the ingroup valuation in *Incentive outgroup*, \$51.31 in *Incentive minimum*, and \$67.98 in *Incentive ingroup*, a significant difference to the allocations in *Ingroup non-social* (each $p < 0.001$, Wilcoxon test). Thus, favoritism reverses when incentives favor the outgroup, intensifies when incentives favor the ingroup, and disappears under the Rawlsian objective.³² The distributions also shift in the predicted direction. In *Incentive outgroup*, the majority of subjects allocate more to the outgroup-WTP, while in *Incentive minimum* the distribution becomes approximately symmetric around the 50:50 split.

Overall, our results speak against the first three alternative explanations. The strong responses to a single change in the payoff function provide evidence against confusion driving our results. Inconsistent with the indifference account, nearly 80% of choices in *Incentive ingroup* and *Incentive outgroup* are interior, rather than the corner solutions implied under indifference. Finally, the reversal toward outgroup-linked valuations in *Incentive outgroup*, together with the disappearance of favoritism in *Incentive minimum*, is difficult to reconcile under the loyalty account.

Design addressing framing concerns: Non-social investment treatment. To address the remaining concern about residual social framing, we designed the *Non-social investment* treatment, which removes all social wording and presents subjects with pure investment choices under uncertainty.

The treatment was run as part of the experiment described in Section 3.3.1. In that experiment, we measure each subject’s perceived interpersonal uncertainty about ingroup and outgroup members and separately estimate how the subject maps objective variation in WTW valuations into perceived interpersonal uncertainty. We use this subject-specific mapping to construct two investment tasks whose uncertainty mirrors the subject’s perceived ingroup-outgroup uncertainty, but without mentioning recipients, groups, or social identity. Specifically, in each investment choice, subjects receive a \$10 endowment to allocate between two accounts. Each account has two possible multipliers, one of which is selected with equal probability. Ex post, the amount invested in each account is multiplied by the realized multiplier for that account and paid to the subject.

Unbeknownst to subjects, the account multipliers are calibrated to match the interpersonal uncertainty that the subject reported about the ingroup and outgroup members. For example, if a subject’s calibrated ingroup spread is (18, 24) and the calibrated outgroup

³¹As the robustness treatments use the gift card environment, we compare the results to the *Ingroup non-social* decisions in the gift card environment, not the pooled decisions. None of our conclusions change if we use the pooled decisions as comparison.

³²Average ingroup-linked allocations are significantly below 50:50 in *Incentive outgroup* ($p < 0.001$) and not significantly different from 50:50 in *Incentive minimum* ($p = 0.15$).

spread is (12, 30), then the two possible multipliers for one account are 1.8 and 2.4, while those for the other account are 1.2 and 3.0. The investment choice is therefore structurally equivalent to choosing between an account with the calibrated uncertainty of the ingroup and an account with the calibrated uncertainty of the outgroup. The task preserves the uncertainty structure of the social decision while removing interpersonal consequences and social framing. We created two investment tasks, one each for the two dimensions of social groups: hobbies/interests and political affiliation. These choices were randomized to appear either before or after the calibrated social allocation decisions within the same session.

Results. In *Non-social investment*, subjects invest more in the lower-uncertainty account in 40% of decisions, choose the 50:50 split in 46%, and invest less in the lower-uncertainty account in 13%. Thus, investment favors the lower-uncertainty account, similar to how social allocations favor the social group with lower uncertainty in *Only group info* decisions. Importantly, at the individual level, the two choices are significantly correlated: subjects who invest more in the lower-uncertainty account in *Non-social investment* also give more to the ingroup in *Only group info* ($r = 0.26, p < 0.001$; see Panel C of Appendix Figure A.3).

Taken together, these results reject the framing concern. Subjects' response to interpersonal uncertainty persists even when social wording and social consequences are removed.

3.5 A discussion on motivated beliefs

A natural concern is that (self-reported) interpersonal uncertainty may reflect *motivated reasoning*: subjects may distort their beliefs in ways that justify behavior they already prefer, such as favoring the ingroup. Under this account, individuals should systematically manipulate their beliefs about the outgroup members' utility – both the mean and the uncertainty – whenever doing so favors their preferred outcome. Moreover, if reported uncertainty primarily serves as an excuse for ingroup favoritism, such distortions should be strongest when subjects make allocation decisions for which the distortion can help justify giving more to the ingroup.

This account generates three predictions. First, subjects should report lower mean valuations for outgroup members than for ingroup members, since this would directly justify allocating more to the ingroup. Second, subjects should report even higher uncertainty about outgroup valuations in treatments where they made a prior *Ingroup social* allocation, than in treatments where they have not. Third, if elevated outgroup uncertainty is primarily driven by motivated belief distortion, it should disappear in the *Ingroup non-social* condition. There, subjects have a monetary incentive to hold accurate beliefs, since correct beliefs about valuations help maximize their own payoff.³³

³³Consistent with this logic, prior work shows that incentives for belief accuracy mitigate motivated reasoning (e.g., Zimmermann, 2020; Sprengholz et al., 2023).

The data do not support these predictions. First, we find no significant differences in mean beliefs between ingroup and outgroup members (Section 3.2). Second, reported uncertainty does not differ significantly depending on whether beliefs are elicited with or without a prior allocation choice (Appendix Section D.2). Third, behavior in *Ingroup non-social* remains consistent with perceived differences in interpersonal uncertainty, even though subjects have monetary incentives to respond to valuations accurately (Section 3.4.2). These findings are difficult to reconcile with the view that reported outgroup uncertainty is merely a motivated distortion used to rationalize ingroup favoritism.

Moreover, our information-choice task (Section 3.1.3) provides an additional test. If subjects use uncertainty as an excuse for favoring the ingroup, they should avoid information about recipients' valuations, since such information would undermine the excuse. Instead, we find the opposite: 81% of subjects choose to learn recipients' WTW, which reduces interpersonal uncertainty, rather than to learn whether recipients belong to the ingroup or the outgroup. This suggests that subjects do not merely use interpersonal uncertainty as an excuse for group-favoring behavior. Rather, a majority of subjects actively seek valuation-relevant information and treat it as more decision-relevant than group membership itself.

4 Extensions to further domains of prosocial behavior

The previous sections focused on ingroup versus outgroup decisions. We now show that interpersonal uncertainty is not specific to that domain. The same logic applies more broadly in situations where decision-makers care about how much others value the outcomes of their choices. We first examine self versus other decisions, where decision-makers may be more certain of their own valuation than another person's valuation.

4.1 Extension to self versus other decisions

Our mechanism naturally extends to tradeoffs between one's own utility and the utility of another person. In such self-versus-other decisions, interpersonal uncertainty predicts an asymmetry: subjects should perceive greater uncertainty about another person's valuation than about their own. If subjects are averse to this uncertainty, the asymmetry can reinforce self-favoring behavior. We test this prediction by adapting our *social/non-social* paradigm to self-versus-other decisions. The experiment includes 120 subjects acting as DMs.

Design. Each subject makes two decisions, a *Self social* decision and a *Self non-social* decision, in randomized order. The *Self social* decision is a standard dictator game: DMs are endowed with $M = \$100$, which they allocate between themselves and another individual. The money is paid out to both parties in the form of Amazon gift card money as in the *Ingroup social* decision.

The *Self non-social* decision removes social consequences while preserving uncertainty about valuations. Subjects again allocate $M = \$100$, but their choice affects only their own payoff, given by:

$$\Pi(x_{self}, x_{other}) = x_{self} \cdot WTP_{self}/100 + x_{other} \cdot WTP_{other}/100$$

where WTP_{self} and WTP_{other} denote the decision-maker's and the matched individual's respective WTP for the gift card, and x_{self} and x_{other} are the amounts allocated to the DM's own WTP and to the matched individual's WTP. Accordingly, DMs are incentivized to maximize the sum of their WTP and the WTP of the matched individual, with both WTPs receiving equal weight. All other elements match the ingroup versus outgroup setting.

In addition, the *Belief measurement* experiment also contained a treatment *Other IU belief* where we elicit perceived uncertainty using the question: "How certain are you about how much [the other person/you yourself] would value Amazon gift card money?" Subjects respond on an 11-point scale. As before, we construct relative interpersonal uncertainty as uncertainty about the other person's valuation minus uncertainty about one's own valuation.

Results. We start with beliefs. In total, 98% of subjects perceive non-zero uncertainty about a randomly selected other person's valuation, and 74% perceive uncertainty about their own valuation. Subjects perceive, on average, 3.11 scale points higher uncertainty for the other person's valuation than for their own ($p < 0.001$, paired Wilcoxon tests).³⁴ In total, 84% of subjects report higher uncertainty about the other person's valuation, 7% report higher uncertainty about their own valuation, and 10% report no difference.³⁵

Moving to the allocation decisions, in the *Self social* decision we replicate the typical pattern of dictator game behavior found in the literature (Engel, 2011): In total, 61% of subjects allocate more money to themselves, 3% allocate more to the other person, and 36% implement the 50/50 split (see Panel A of Figure A.7 for the distribution). On average, subjects allocate \$69.05 to themselves, significantly more than the equal split ($p < 0.001$, one-sample Wilcoxon test). In *Self non-social*, subjects allocate on average \$64.12 to their own WTP, again a significant deviation from the equal split ($p < 0.001$, one-sample Wilcoxon test). As Panel B of Figure A.7 shows, the distribution is also similarly shaped to the *Self social* case. In total, 58% of subjects allocate more money to their own WTP, 17% allocate more to the other individual's WTP, and 25% implement the 50/50 split. Allocations in the *Self non-social* setting thus closely replicate the behavior we observe in *Self social*.³⁶

³⁴We also elicited perceived uncertainty in the *Self social* treatment itself after the allocation decisions. We find similar results, with subjects reporting, on average, 2.59 scale points higher uncertainty for the other person's valuation.

³⁵We also elicited participants' beliefs about others' average valuation, which we can compare with their own stated valuation. On average, participants believed others had a WTP of \$83.43, while their own elicited WTP was \$86.69 on average. In total, 41% believed others had a higher WTP than they stated themselves, 7% believed it was the same, and 53% believed others had a lower WTP.

³⁶To show that people are attentive to the *non-social* incentives in the self versus other setting, we use the same incentive treatment as in the ingroup versus outgroup setting (Section 3.4.3) and apply it to the current

Moreover, we find a strong within-subject association: the correlation between the *Self social* and *Self non-social* decisions is $r = 0.71$ ($p < 0.001$); see Panel C of Appendix Figure A.7 for the corresponding binscatter plot. Hence, the *Self non-social* decision strongly predicts the *Self social* decision.³⁷

4.2 Extension to behavioral interventions

Our results so far show that interpersonal uncertainty predicts prosocial behavior across recipients within a fixed decision environment. We next ask whether it also helps explain changes in behavior caused by interventions. We focus on two prominent interventions in the prosocial-behavior literature: the *identifiable victim effect* and the *contact effect*. The identifiable victim effect describes the increase in giving when information about a specific, identified individual is provided, compared to general information about a large group (see Lee and Feeley, 2016, for a meta-analysis). The contact effect describes the empirical finding that individuals exhibit greater prosocial behavior toward others after engaging in some form of social interaction (see Clochard, 2024; Lowe, 2025, for recent meta-analyses).

We hypothesize that identifiable victim interventions reduce interpersonal uncertainty by providing concrete details about an individual, making their needs and circumstances easier to imagine with greater confidence. Similarly, social interactions in contact interventions offer insights into others' lives and experiences, reducing uncertainty about their preferences and circumstances. We further hypothesize that this reduction in uncertainty *mediates* higher prosocial actions in both environments.

Design. Each intervention features a treatment and a control condition. In the two *Contact* conditions, subjects are matched with another individual. In the *Contact control* condition, subjects then play a dictator game, allocating \$100 between themselves and their matched individual. In *Contact treatment*, subjects first complete a conversational exchange: they choose and answer three of five ice-breaker questions, and then see their matched partner's answers to the same questions.

In the two identifiable victim (IV) conditions, subjects allocate \$100 between themselves and recipients living in Malawi through a donation to the charity GiveDirectly. In both the control and treatment conditions, subjects first receive a general introduction about the charity and Malawi. In the *IV control* condition, they then receive some general information about the recipients in Malawi. In *IV treatment*, they receive similar informational content but through a first-hand account from a specific, identified Malawi recipient who shared their story on the GiveDirectly website.

setting. Our results are similar: subjects understand the incentives and react to them as hypothesized. We provide the results in Appendix Section E.5.

³⁷In addition, perceived interpersonal uncertainty (Section 3.2) is significantly correlated with *Self non-social* decisions ($r = 0.24$ ($p < 0.01$)), see Panel B of Appendix Figure A.4.

In each condition, we elicit subjects' interpersonal uncertainty after their allocation decision. Specifically, in the contact condition, we elicit interpersonal uncertainty about their matched partner and in the identifiable victim conditions about the recipients in Malawi. In total, 602 subjects participated in the experiment, 149 each in *Contact treatment* and *Contact control*, and 152 each in *IV treatment* and *IV control*.

Results. First, we replicate the effectiveness of the interventions in increasing prosocial behavior. While subjects in *Contact control* allocate \$69.04 to themselves, in *Contact treatment* giving to self decreases to \$53.48 ($p < 0.001$, two-sample Wilcoxon test). In *IV control*, subjects allocate \$56.70 to themselves, which decreases to \$48.74 in *IV treatment* ($p = 0.046$, two-sample Wilcoxon test).

Both interventions also reduce interpersonal uncertainty. Average uncertainty about the matched partner falls from 3.22 in *Contact control* to 2.68 in *Contact treatment* ($p = 0.034$). Average uncertainty about recipients in Malawi falls from 3.14 in *IV control* to 1.97 in *IV treatment* ($p < 0.001$). Moreover, interpersonal uncertainty predicts self-favoritism in both settings, as shown in Appendix Figure A.5 (identifiable victim: $r = 0.37$, $p < 0.001$; contact: $r = 0.33$, $p < 0.001$). Thus, both interventions increase giving and reduce uncertainty, consistent with interpersonal uncertainty helping explain why these interventions affect prosocial behavior.

4.2.1 Interpersonal uncertainty as mediator for behavioral interventions

We next test if the behavioral effect of each intervention can be *fully explained* by its effect on interpersonal uncertainty. To do so, we apply the test of the sharp null of full mediation developed by Kwon and Roth (2026). The test asks whether the joint distribution of treatment status, interpersonal uncertainty, and giving is consistent with the hypothesis that treatment affects giving only through interpersonal uncertainty. Crucially, unlike standard mediation analysis, this does not require any assumptions about the quasi-random assignment of interpersonal uncertainty.

Description of mediation test. For each intervention, let $D_i \in \{0, 1\}$ denote treatment assignment. Let $M_i \in \{0, 1\}$ denote the binary mediator, where $M_i = 1$ indicates low interpersonal uncertainty and $M_i = 0$ indicates high interpersonal uncertainty. Let Y_i denote prosocial behavior, measured as giving to the other recipient. The sharp null of full mediation is

$$H_0 : Y_i(1, M_i) = Y_i(0, M_i) \quad \text{for all } M_i \in \{0, 1\}.$$

That is, once interpersonal uncertainty is held fixed, the intervention has no remaining effect on giving. The test relies on two assumptions (Kwon and Roth, 2026). First, that the treatment is randomly assigned, which holds by design. Second, that the treatment weakly reduces interpersonal uncertainty: since $M_i = 1$ denotes low uncertainty, $M_i(1) \geq M_i(0)$.

To implement the test, we discretize interpersonal uncertainty into two bins, high and low, and discretize allocation-to-self into four equal bins.³⁸ For every allocation-to-self bin A , the sharp null then has two testable implications: First, the treatment should not increase the joint probability of allocating in A while being in the high-uncertainty group. Such an increase would imply that the treatment affects allocations even among subjects whose interpersonal uncertainty is always high. Second, as the treatment can only add compliers to the low-uncertainty group, the treatment group should not have a lower joint probability of allocating in A while being in the low-uncertainty group.

Results of mediation test. Appendix Figure A.6 plots the empirical joint distributions relevant to the testable implications mentioned above. For the identifiable-victim treatment, the empirical distributions are consistent with the inequalities implied by full mediation: relative to control, the treatment shifts subjects toward lower interpersonal uncertainty and higher giving in a way that can be rationalized exclusively by movement through the interpersonal-uncertainty channel. Correspondingly, the formal test does not reject the sharp null that interpersonal uncertainty fully mediates the identifiable-victim effect ($p = 0.89$). For the contact treatment, the empirical distributions violate the inequalities implied by full mediation, and full mediation is rejected ($p < 0.001$).

Taken together, these results show that the two interventions operate differently. For the identifiable-victim treatment, the data are statistically consistent with interpersonal uncertainty being a complete mediator. For the contact treatment, they are not.

5 Conclusion

In this paper, we provide a conceptual framework and implement a series of experiments documenting how interpersonal uncertainty influences prosocial behavior. We show that a significant degree of heterogeneity in prosocial behavior, both within and across decision settings, is driven by people’s differential responses to interpersonal uncertainty. As a consequence, precise identification of social preferences from prosocial behavior requires explicit accounting for interpersonal uncertainty. We demonstrate how a researcher can exogenously vary interpersonal uncertainty to explicitly measure and control for it.

Finally, our framework supports the idea that prosocial behavior is malleable. Under the assumption that "exposure reduces interpersonal uncertainty", it helps explain the dynamics of prosocial behavior in response to intergroup contact created by spatial proximity (Bursztyn et al., 2024), shared classes (Rao, 2019), shared living (Corno, La Ferrara, and Burns, 2022), sports events (Mousa, 2020; Lowe, 2021) and attending youth camps (Ghosh

³⁸Discretization ensures that the joint bins of treatment status, mediator value, and allocation-to-self category contain enough observations. Our results are robust to alternative numbers of allocation-to-self bins, e.g., two or eight bins.

et al., 2026). Indeed, we provide evidence that a stylized contact intervention not only increases prosocial behavior, but also decreases interpersonal uncertainty. Similarly, our conceptual framework and results clarify how people’s degree of favoritism towards specific groups varies based on their closeness (Fong and Luttmer, 2009), salience of shared experiences (McLeish and Oxoby, 2011), or (perceived) similarity (Goeree et al., 2010) to ingroup members. Overall, our results suggest that targeting and reducing interpersonal uncertainty could foster prosocial behavior, bridge animosities, and decrease intergroup conflict.

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Appendix: Proofs

Proof of Theorem 1. For simplicity, whenever possible we will write $E_{v_i \sim f_i}$ simply as E_{f_i} . From the utility expression, we get

$$\begin{aligned}\frac{d}{dx}EU(x) &= E_{f_1, f_2}((v_1 - v_2)U'(v_1x + v_2(100 - x))) \\ \frac{d^2}{dx^2}EU(x) &= E_{f_1, f_2}((v_1 - v_2)^2U''(v_1x + v_2(100 - x))) < 0\end{aligned}$$

as $U'' < 0$ and $f_1, f_2 \geq 0$. $\frac{d^2}{dx^2}EU(x)$ being strictly negative implies that $\frac{d}{dx}EU(x) = 0$ must be obtained at a unique point. Evaluating the first derivative at $x = 50$, we get

$$\frac{d}{dx}EU(x)|_{x=50} = E_{f_1, f_2}(v_1 - v_2)U'(50v_1 + 50v_2) \quad (2)$$

(i) When f_1 and f_2 are identical, then we can also rewrite

$$\begin{aligned}\frac{d}{dx}EU(x) &= E_{v_1 \sim f_2}E_{v_2 \sim f_1}(v_1 - v_2)U'(v_1x + v_2(100 - x)) \\ &= E_{v_2 \sim f_2}E_{v_1 \sim f_1}(v_2 - v_1)U'(v_2x + v_1(100 - x)) \\ &= E_{f_1, f_2}(v_2 - v_1)U'(v_2x + v_1(100 - x))\end{aligned}$$

where the first step integrates v_1 over f_2 and v_2 over f_1 instead, the second step interchanges the names of the variables of integration (v_1 and v_2). The last step interchanges the order of integration. Evaluating the final expression at $x = 50$, we get

$$\frac{d}{dx}EU(x)|_{x=50} = E_{f_1, f_2}(v_2 - v_1)U'(50v_1 + 50v_2) \quad (3)$$

Equations 2 and 3 together imply $\frac{d}{dx}EU(x)|_{x=50} = -\frac{d}{dx}EU(x)|_{x=50} = 0$.

(ii) When f_2 is a mean-preserving spread of f_1 , then there exists a random variable $z \sim f_z$ with zero expectation conditional on any given value of v_1 , such that v_2 has the same distribution as $v_1 + z$, or in other words, $v_2 \stackrel{d}{=} v_1 + z$. Therefore, we can replace v_2 by a variable $w_1 + z$ where w_1 and v_1 both have identical distribution f_1 .

$$\frac{d}{dx}EU(x) = E_{v_1 \sim f_1}E_{w_1 \sim f_1}E_{z|v_1, w_1}(v_1 - w_1 - z)U'(v_1x + (w_1 + z)(100 - x)) \quad (4)$$

$$\frac{d}{dx}EU(x)|_{x=50} = E_{f_1}E_{f_1}E_{z|v_1, w_1}(v_1 - w_1 - z)U'(50v_1 + 50w_1 + 50z) \quad (5)$$

Because $E_{f_1} E_{f_1}$ integrates with respect to two identical independent distributions, we can interchange their variable names (w_1 and v_1) in Equation 5:

$$\frac{d}{dx} EU(x)|_{x=50} = E_{f_1, f_1} E_{z|v_1, w_1} (w_1 - v_1 - z) U' (50v_1 + 50w_1 + 50z) \quad (6)$$

Adding equations 5 and 6, and then using the law of iterated expectations:

$$\begin{aligned} 2 \frac{d}{dx} EU(x)|_{x=50} &= -E_{f_1, f_1} (E_{z|v_1, w_1} 2z U' (50v_1 + 50w_1 + 50z)) \\ &> -E_{f_1, f_1} (E_{z|v_1, w_1} 2z U' (50v_1 + 50w_1)) \\ &= -E_{f_1, f_1} U' (50v_1 + 50w_1) (E_{z|v_1, w_1} 2z f_z(z|v_1) dz) \\ &= 0 \end{aligned}$$

The inequality uses the fact: $z U' (50v_1 + 50w_1 + 50z) < z U' (50v_1 + 50w_1)$ for both $z > 0$ and $z < 0$. The last step follows from the fact that $E_{z|v_1, w_1} z = 0$. Therefore, $\frac{d}{dx} EU(x)|_{x=50} > 0$, and thus, the optimal allocation $x^* > 50$. Next,

$$\begin{aligned} \frac{d}{dx} EU(x)|_{x=100} &= E_{f_1} E_{f_1} E_{z|v_1, w_1} (v_1 - w_1 - z) U' (100v_1) \\ &= E_{f_1} E_{f_1} U' (100v_1) E_{z|v_1, w_1} (v_1 - w_1 - z) \\ &= E_{f_1} E_{f_1} U' (100v_1) (v_1 - w_1) \\ &= E_{f_1} E_{f_1} U' (100w_1) (w_1 - v_1) \\ &= \frac{1}{2} E_{f_1} E_{f_1} [U' (100w_1) (w_1 - v_1) + U' (100v_1) (v_1 - w_1)] \\ &= \frac{1}{2} E_{f_1} E_{f_1} (U' (100w_1) - U' (100v_1)) (w_1 - v_1) \\ &< 0 \end{aligned}$$

Step 1 substitutes $x = 100$ into the expression of $\frac{d}{dx} EU(x)$ derived at the beginning of the proof. Step 2 uses that $U' (100v_1)$ is independent of z . Step 3 uses $E_{z|v_1, w_1} z = 0$. Step 4 uses the property that v_1, w_1 are drawn i.i.d. from f_1 , and hence those two variable names can be interchanged. Step 5 uses the average of the two expressions from the previous lines. The last step uses the property that U' is decreasing.

As $\frac{d}{dx} EU(x)|_{x=100} < 0$, the concavity of the expression implies that $\frac{d}{dx} EU(x) = 0$ must be obtained at some $50 < x < 100$.

(iii) The first derivative of the objective function, evaluated at x^* , should be zero.

$$E_{f_1, f_2} (v_1 - v_2 - c) U' (x^* v_1 + (100 - x^*) (v_2 + c)) = 0 \quad (7)$$

First, taking the implicit derivative of the last equation w.r.t. c , and then rearranging:

$$E_{f_1, f_2} [-U' + (v_1 - v_2 - c)^2 \frac{dx^*}{dc} U'' + (v_1 - v_2 - c) (100 - x^*) U''] = 0$$

Next, we rearrange and then bound $\frac{dx^*}{dc}$ in 6 steps as explained below. Under CARA,

$$\begin{aligned}
\frac{dx^*}{dc} &= \frac{E_{f_1, f_2} - U'}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} + \frac{E_{f_1, f_2}(v_1 - v_2 - c)(100 - x^*)U''}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} \\
&= \frac{E_{f_1, f_2} - U'}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} + \frac{E_{f_1, f_2}(v_1 - v_2 - c)(100 - x^*) \times \frac{U''_{100}}{U'_{100}} U'}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} \\
&= \frac{E_{f_1, f_2} - U'}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} + \frac{(U''_{100})(100 - x^*)}{U'_{100}} \frac{E_{f_1, f_2}(v_1 - v_2 - c)U'}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} \\
&= \frac{E_{f_1, f_2} - U'}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} + \frac{(U''_{100})(100 - x^*)}{U'_{100}} \times \frac{0}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} < 0
\end{aligned}$$

The second step utilizes the assumption of constant absolute risk aversion: $\frac{U''}{U'} = \frac{U''_{100}}{U'_{100}}$, and hence, $U'' = \frac{U''_{100}}{U'_{100}} U'$. The third step simply reorganizes the numerator in the second additive term. The fourth step uses equation 7 to set $E_{f_1, f_2}(v_1 - v_2 - c)U'$ to zero. The last step uses $U' > 0, U'' < 0$.

Under CRRA preferences,

$$\begin{aligned}
\frac{dx^*}{dc} &= \frac{E_{f_1, f_2}[-U' + (v_1 - v_2 - c)(100 - x^*)U'']}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} \\
&= \frac{E_{f_1, f_2}[-U' - (x^* v_1 + (100 - x^*)(v_2 + c))U'' + 100v_1 U'']}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} \\
&= \frac{E_{f_1, f_2}[-U' + rU' + 100v_1 U'']}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''} \\
&= \frac{E_{f_1, f_2}[-(1 - r)U' + 100v_1 U'']}{-E_{f_1, f_2}(v_1 + c - v_2)^2 U''}
\end{aligned}$$

The third step utilizes $r < 1$. In the last expression, the numerator is negative as $r < 1, v_1 \geq 0, U'' < 0$ and the denominator is positive, which concludes the proof. $\square$

ONLINE APPENDIX

A Additional tables and figures

Table A.1: Overview of experiments

Paradigm	Treatments/Decisions	N	Purpose
Ingroup/outgroup			
Synthetic group paradigm	<i>Uncertainty without group information, Group information without uncertainty, High uncertainty on ingroup members, & High uncertainty on outgroup members</i>	120	Measures aversion to interpersonal uncertainty and its causal effect on ingroup favoritism.
Belief measurement	<i>Ingroup IU belief & Other IU belief*</i>	120	Measures perceived interpersonal uncertainty.
Synthetic group calibration	<i>Only group info, Only WTW info, WTW + group info, & Non-social investment</i>	120	Quantifies the importance of interpersonal uncertainty relative to group preferences.
Ingroup social/non-social	<i>Ingroup social & Ingroup non-social</i>	119	Tests how removing the social component of allocation decisions while preserving uncertainty influences choices.
	<i>Ingroup effort social & Ingroup effort non-social</i>	121	
Ingroup social/non-social robustness	<i>Incentive ingroup & Incentive outgroup</i>	120	Tests for confounds in the social/non-social paradigm.
	<i>Incentive minimum</i>	62	
	<i>Non-social investment</i>	**	
Extensions			
Self versus other social/non-social	<i>Self social & Self non-social</i>	120	Extends the social/non-social paradigm to the dictator game.
Self versus other social/non-social robustness	<i>Incentive self & Incentive other</i>	120	Tests for confounds in the self versus other experiment.
Behavioral interventions	<i>Contact control, Contact treatment, IV control & IV treatment*</i>	602	Tests whether behavioral interventions change interpersonal uncertainty.

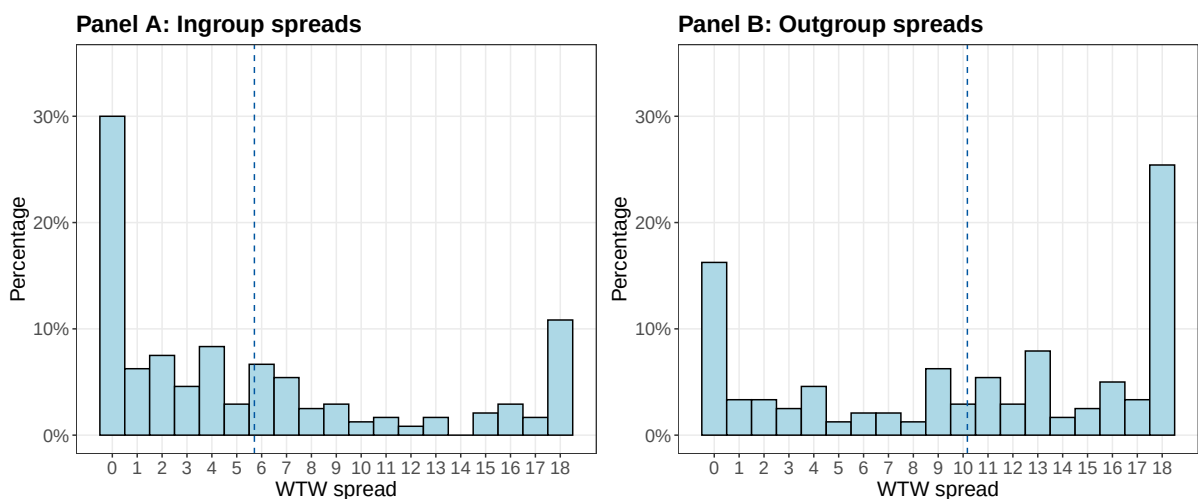
Notes: *Treatments were implemented between subjects. **The *Non-social investment* treatment was implemented together with the synthetic group calibration experiment. All experiments were pre-registered; see Appendix Section F. See Appendix Section F for further details and the mapping between preregistration and paper.

Table A.2: The influence of changes in interpersonal uncertainty on ingroup favoritism

	<i>Dependent variable: Allocation to ingroup</i>					
	Both groups pooled		Political views		Hobbies/interests	
	(1)	(2)	(3)	(4)	(5)	(6)
Constant (Ingroup pref.)	0.666*** (0.104)	0.741*** (0.132)	0.964*** (0.140)	0.898*** (0.186)	0.367*** (0.101)	0.584*** (0.130)
Interpersonal unc.	0.742*** (0.115)		0.720*** (0.121)		0.764*** (0.127)	
Higher unc. ingroup		−0.856*** (0.144)		−0.622*** (0.178)		−1.090*** (0.178)
Higher unc. outgroup		0.629*** (0.135)		0.819*** (0.187)		0.438*** (0.154)
Subjects	120	120	120	120	120	120
Observations	720	720	360	360	360	360
R ²	0.095	0.096	0.080	0.081	0.121	0.129

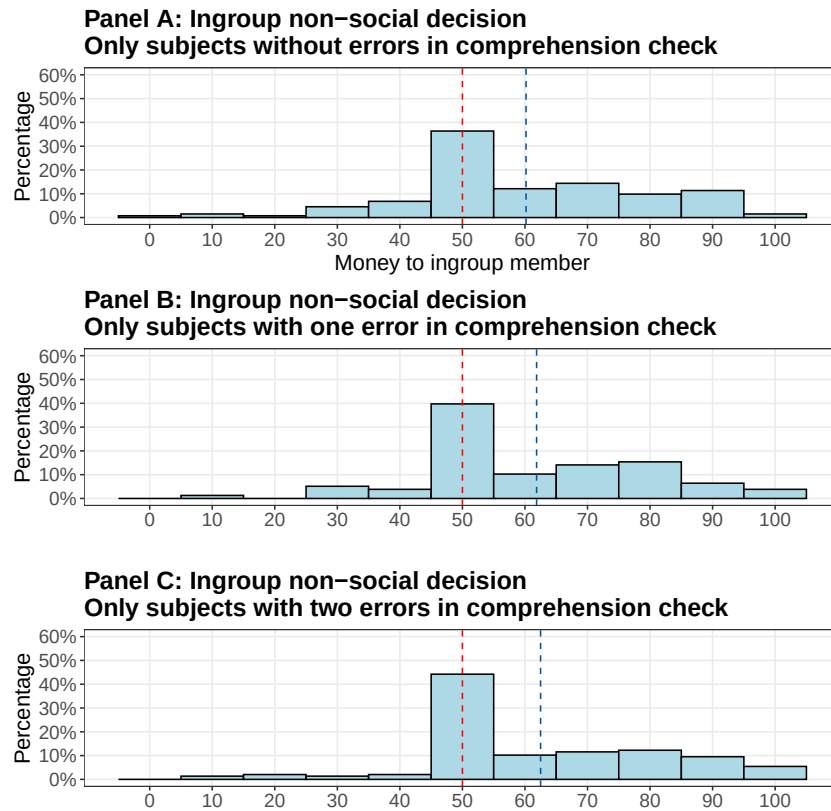
Notes: The table shows OLS estimates. The dependent variable is the amount allocated to the ingroup (out of \$10), subtracted by five. “Interpersonal uncertainty” is equal to 1, 0, or -1 when the decision has *High uncertainty on ingroup members*, *Group information without uncertainty*, and *High uncertainty on outgroup members*, respectively. “Higher uncertainty ingroup” is equal to 1 when the decision has *High uncertainty on ingroup members* (*High uncertainty on outgroup members*) and zero otherwise. In columns (1) and (2), decisions are pooled across both social groups (political views and hobbies/interests) that define the ingroup and outgroup. In columns (3) and (4), only decisions involving groups based on political views are considered, while in columns (5) and (6) only decisions involving groups based on hobbies/interests are considered. Standard errors (in parentheses) are clustered at the subject level. Significance levels: * $p < 0.1$, ** $p < 0.05$ and *** $p < 0.01$.

Figure A.1: Implemented WTW spreads for the *Only WTW info* and *WTW + Group info* decisions



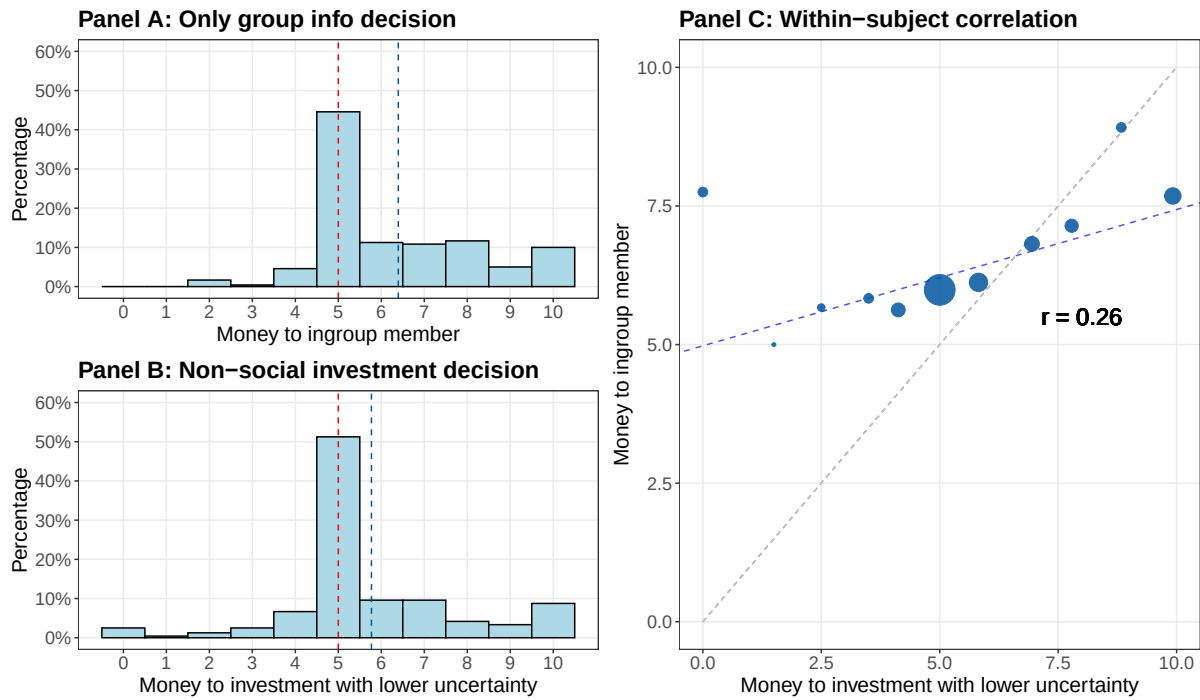
Notes: Histogram of the implemented ingroup (Panel A) and outgroup (Panel B) WTW spreads for the *Only WTW info* and *WTW + Group info* decisions. WTW spread is defined as the difference in WTWs between the two members of the synthetic groups. The blue line denotes the average spread. Spreads are pooled across two social groups (shared hobbies/interests and political views), displaying $n = 240$ spreads in each panel.

Figure A.2: Ingroup non-social decisions sorted according to comprehension question errors



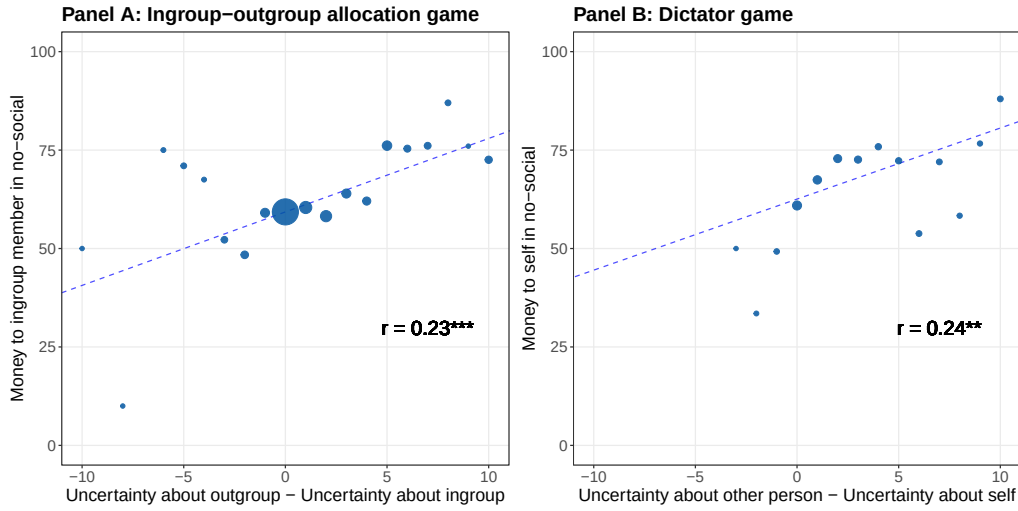
Notes: Histogram of *Ingroup non-social* decisions. The x-axis denotes the amount of money (out of \$100) allocated to the ingroup member instead of the outgroup member. The red line denotes the even split, the blue line the average allocation. The decisions have consequences only for the decision-makers, with their payoff depending on the ingroup and outgroup members' valuations of money. In all panels, the binwidth is 10. Decisions are pooled across the three groups (shared hobbies/interests, political views, and religious beliefs).

Figure A.3: Non-social investment decision robustness treatment result



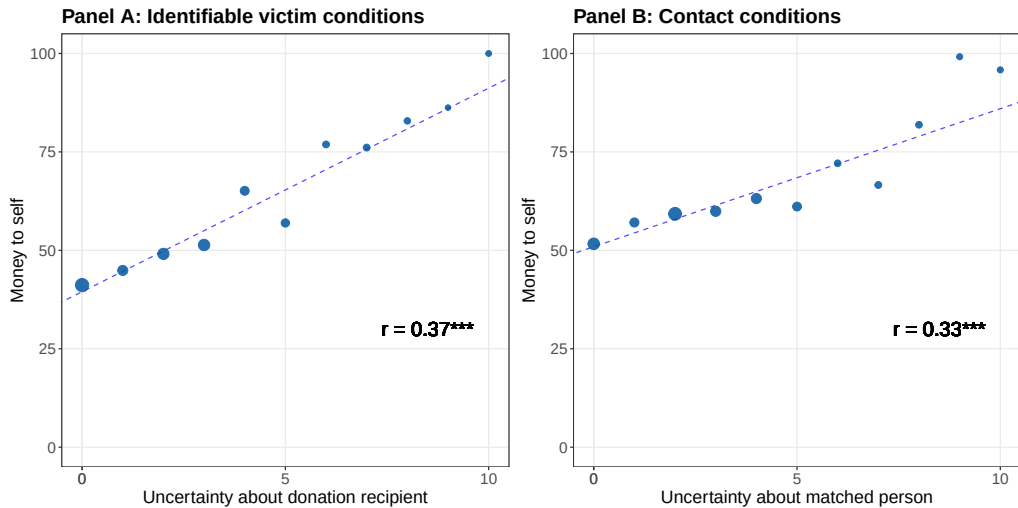
Notes: **Panels A and B:** Histograms of the *Only group info* (Panel A) and *Non-social investment* (Panel B) decisions. The binwidth is 1. In Panel A, the x-axis denotes the amount of money (out of \$10) that subjects allocated to the ingroup member instead of the outgroup member. In Panel B, the x-axis denotes the amount (out of \$10) that subjects allocate to the investment account with the lower uncertainty instead of the one with the higher uncertainty. The red line denotes the even split, the blue line the average allocation. **Panel C:** Binned scatter plot of the *Only group info* and *Non-social investment* decisions. We divide *Non-social investment* choices into 10 bins and plot the average giving to self in *Only group info* for each bin. The blue line displays the linear fit of regressing the *Only group info* on the *Non-social investment* decision. The correlation coefficient is $r = 0.26$. Displayed are $n = 240$ decisions by 120 subjects.

Figure A.4: Association between reported interpersonal uncertainty and non-social allocation decisions



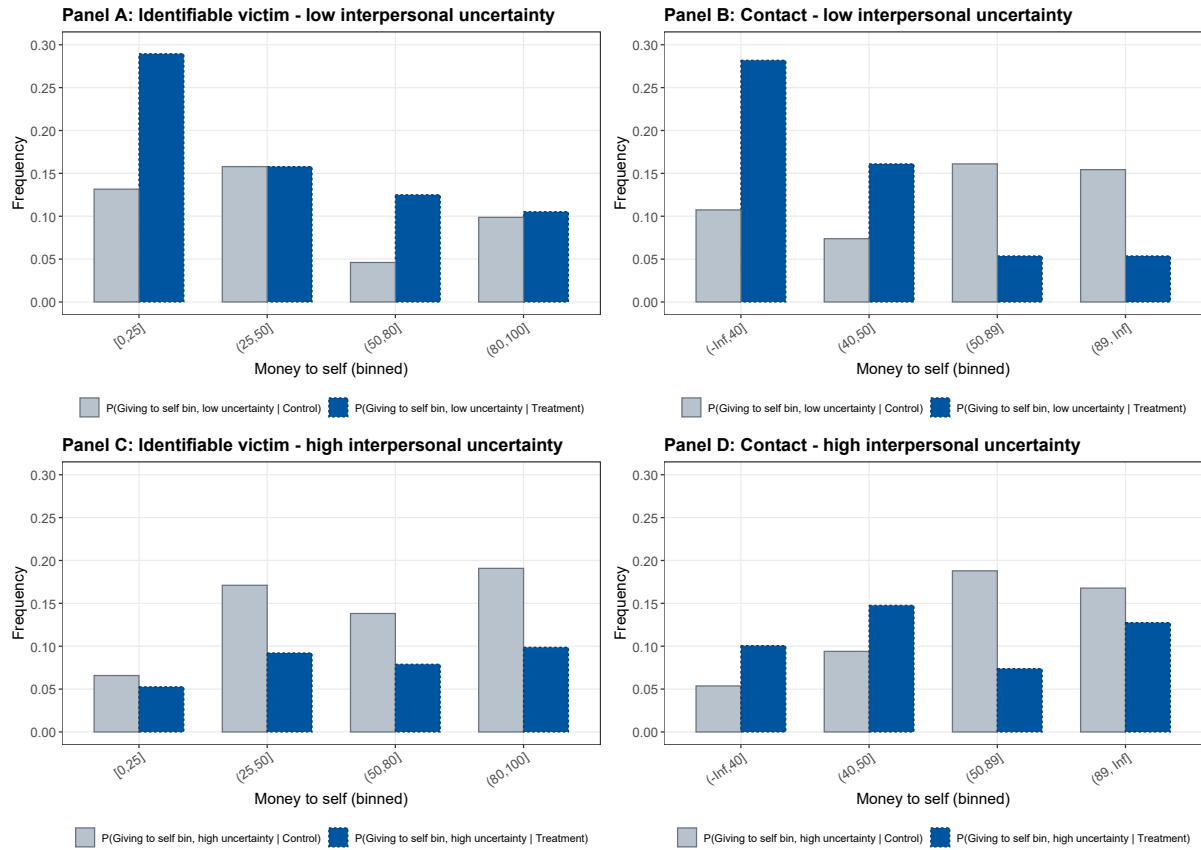
Notes: Binned scatter plots showing the relationship between reported interpersonal uncertainty and non-social decision-making in ingroup versus outgroup allocation games (Panel A) and the dictator game (Panel B). **Panel A:** The x-axis shows the difference in interpersonal uncertainty between ingroup and outgroup members, and the y-axis shows money allocated to the ingroup member in the *Ingroup non-social* decision. $n = 720$ decisions from 240 subjects, pooled across three social groups and two environments (*Gift card* and *Effort*, see Section 3.4). **Panel B:** The x-axis shows the difference in interpersonal uncertainty between the dictator game recipient and oneself, and the y-axis shows the amount allocated to self in the *Self non-social* decision. $n = 120$ decisions from 120 subjects. In both panels, we divide the interpersonal uncertainty measure into 20 bins and plot the average allocation in each bin. The linear regression fit is shown with the blue dotted line, and correlation coefficients are displayed. Significance levels: * $p < 0.1$, ** $p < 0.05$ and *** $p < 0.01$.

Figure A.5: Association between reported interpersonal uncertainty and allocation decisions



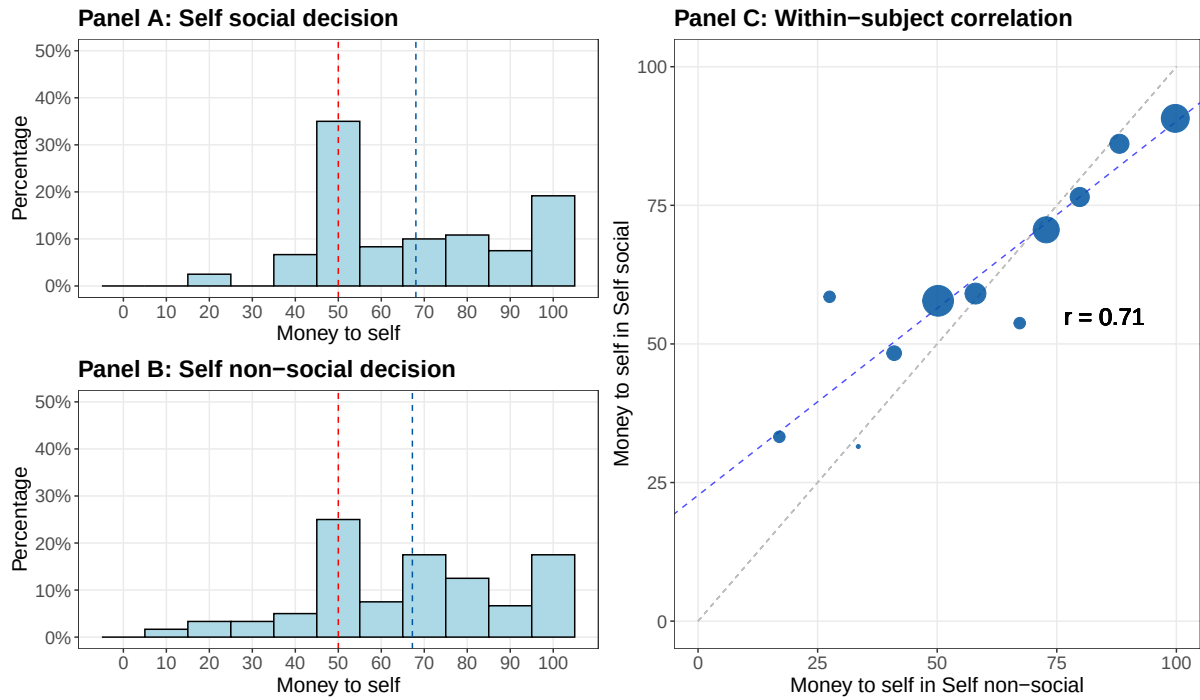
Notes: Binned scatter plots showing the relationship between reported interpersonal uncertainty and decision-making for the *Identifiable victim* intervention (Panel A) and the *Contact* intervention (Panel B). **Panel A:** The x-axis shows uncertainty about the recipients of the GiveDirectly donation, and the y-axis shows money allocated to oneself. $n = 304$ decisions from 304 subjects pooled across treatment and control groups. **Panel B:** The x-axis shows uncertainty about the matched person, and the y-axis shows money allocated to oneself. $n = 298$ decisions from 298 subjects pooled across treatment and control groups. In both panels, we divide the interpersonal uncertainty measure into 10 bins and plot the average allocation in each bin. The linear regression fit is shown with the blue dotted line, and correlation coefficients are displayed. Significance levels: * $p < 0.1$, ** $p < 0.05$ and *** $p < 0.01$.

Figure A.6: Partial density plots of giving and interpersonal uncertainty in the behavioral interventions treatments



Notes: Partial density plots of the behavioral interventions. “Money to self” denotes the amount subjects allocate to themselves instead of recipients in Malawi (*Identifiable victim*) or the matched partner (*Contact*). Giving is discretized into four bins with an equal number of subjects in each. Interpersonal uncertainty is median-split into low and high uncertainty, where low uncertainty is defined as reported uncertainty weakly below the sample median. **Panels A and C** show the results from the *Identifiable victim* intervention. **Panels B and D** show the results from the *Contact* intervention. Grey bars denote the control condition in each intervention, blue bars the treatment. Sample sizes are 304 subjects for the *Identifiable victim* intervention and 298 subjects for the *Contact* intervention.

Figure A.7: Main results self versus other decision



Notes: **Panels A and B:** Histograms of the *Self social* (Panel A) and *Self non-social* (Panel B) decisions. The binwidth is 10. In Panel A, the x-axis denotes the amount of money (out of \$100) that subjects allocate to themselves instead of another individual, with decisions having consequences for them and the other individual. In Panel B, the x-axis denotes the amount (out of \$100) that subjects allocate to their own valuation instead of another individual's valuation, with their payoff depending on their own and the other individual's valuation of money. The red line denotes the even split, the blue line the average allocation. **Panel C:** Binned scatter plot of the *Self social* and *Self non-social* decisions. We divide *Self non-social* choices into 10 bins and plot the average giving to self in *Self social* for each bin. The blue line displays the linear fit of regressing the *Self social* on the *Self non-social* decision. The correlation coefficient is $r = 0.71$. Displayed are $n = 61$ (Panel A), $n = 59$ (Panel B), and $n = 120$ (Panel C) decisions.

B Rawlsian preferences under interpersonal uncertainty

In Section 2, we showed that utilitarianism generates patterns of prosocial behavior given certain assumptions on interpersonal uncertainty. This raises the question of whether every commonly used welfare criterion delivers similar patterns under the right parameters given our assumptions. Here, we show that Rawlsian preferences – one of the most discussed welfare criteria – are insensitive to interpersonal uncertainty. Under Rawlsian preferences, only the utility of the least well-off recipient matters. In our context, Rawlsian preferences mean the utility that individual i receives from allocating x to the ingroup member and $(100 - x)$ to the outgroup member is $u_{RAWLS} = \min\{v_1x, v_2(100 - x)\}$. As Theorem 2 shows, a DM will then split the money equally independently of interpersonal uncertainty.

Theorem 2. *Suppose individual i has Rawlsian preferences. Then irrespective of i 's risk attitude ($U'' \leq 0$ or $U'' \geq 0$), her optimal allocation is $x^* = 50$, in both the following cases, i) $f_1 = f_2$, and, ii) f_2 is a mean-preserving spread of f_1 .*

Proof of Theorem 2. As v_1, v_2 are random variables, i 's expected utility from allocating x to the outgroup is:

$$EU(x) = E_{v_1 \sim f_1, v_2 \sim f_2} \min\{v_1x, v_2(100 - x)\}$$

For any $x \in [0, 50) \cup (50, 100]$,

$$\min\{v_1(100 - x), v_2x\} + \min\{v_1x, v_2(100 - x)\} \leq v_1(100 - x) + v_1x = 100v_1$$

with strict inequality whenever $v_1 \neq v_2$.³⁹

Similarly, $\min\{v_1(100 - x), v_2x\} + \min\{v_1x, v_2(100 - x)\} \leq 100v_2$ with strict inequality whenever $v_1 \neq v_2$. Putting these two inequalities together, we get

$$\min\{v_1(100 - x), v_2x\} + \min\{v_1x, v_2(100 - x)\} \leq \min\{100v_1, 100v_2\}$$

with strict inequality whenever $v_1 \neq v_2$. Next, using $f_1 = f_2$,

$$\begin{aligned} EU(x) &= E_{v_1 \sim f_1, v_2 \sim f_2} \min\{v_1x, v_2(100 - x)\} \\ &= E_{v_1 \sim f_1, v_2 \sim f_2} \min\{v_1(100 - x), v_2x\} \end{aligned}$$

Therefore, for any $x \in [0, 50) \cup (50, 100]$,

$$\begin{aligned} EU(x) &= \frac{1}{2} \times E_{v_1 \sim f_1, v_2 \sim f_2} (\min\{v_1x, v_2(100 - x)\} + \min\{v_1(100 - x), v_2x\}) \\ &< \frac{1}{2} \times E_{v_1 \sim f_1, v_2 \sim f_2} \min\{100v_1, 100v_2\} \\ &= E_{v_1 \sim f_1, v_2 \sim f_2} \min\{50v_1, 50v_2\} \end{aligned}$$

The first inequality becomes strict as $v_1 \neq v_2$ with positive probability in the integration. This proves part (i), and a similar proof works for part (ii) after v_2 is replaced with $w_1 + z_1$, as in the proof of Theorem 1. □

³⁹If $x = 50$, then strict inequality does not hold under $v_1 < v_2$.

C Extensions and further details on the synthetic group paradigm and experiment

C.1 Interpretation of willingness-to-work differences

Elicitation. We included an open-ended question to capture how subjects interpret differences in our willingness-to-work (WTW) measure, which denotes the number of tasks for which subjects are willing to work to receive a \$10 bonus. Specifically, at the end of *Ingroup social* and *Ingroup non-social* experiments in the *Effort environment* – after subjects had completed the WTW elicitation and allocation decisions – they answered the following question:

Why might some Prolific participants be willing to complete more tasks than other participants in order to receive the \$10 bonus payment? Please explain in one to three sentences.

We used the large language model GPT-5.5 to categorize the open-ended responses from 121 subjects. Responses could receive multiple codes. We order the categories by their relevance for interpreting WTW as information about the marginal utility of money. See Table C.1 for a description and a representative example of each category. Table C.2 then provides the frequencies for each category. The first category, financial need, most directly reflects wealth heterogeneity: 55 subjects (45.5%) explicitly state that some participants may be more willing to complete tasks because they need the money more. The second category captures heterogeneous valuation of money: 45 subjects (37.2%) mention that some participants value the \$10 bonus more, are more money-focused, or are more motivated by earnings, even without explicitly referring to hardship. Overall, 71.1% of subjects mention either financial need, the subjective value of money, or both, as one of the sources of differential WTW.

The remaining categories are progressively less related to the marginal utility of money. Some subjects frame the decision as a time-money tradeoff, which is consistent with the idea that those with higher WTW care relatively more about money, but also admits the alternative interpretation that they care equally about money and have lower costs of effort. In total, 31 subjects mention this category. Accordingly, a cumulative fraction of 84.3% of subjects mention one of the first four categories, which are the categories most closely connected to interpreting WTW as informative about the value of monetary payoffs. The remaining responses point elsewhere: subjects discuss how much other subjects enjoy the task (33 subjects; 27.3%), how much time they have to complete the task (25 subjects; 20.7%), or provide responses that are unclear or unrelated (9 subjects; 7.4%).

Table C.1: Open-ended explanations for willingness to complete more tasks for a \$10 bonus

Category	Description	Representative example
Financial need	Participants may be willing to do more tasks because they need the money more, including financial hardship, bills, necessities, low income, or reliance on Prolific income.	"I think for some people having a higher need for money will motivate them to do more tasks. If they are in more need of the money they may be willing to do it more than someone who does not really need the money."
Subjective value of money	Participants may be more willing because they personally value earning \$10 more, are more money-focused, or are more motivated about current or future earnings (getting invited to more studies), even without explicit financial hardship.	"Some participants may value \$10 more than others. Some may actually rely on the money they receive as a bonus for bills or food. Some may just see this as a side hustle and want the \$10 to toss into their savings."
Time-money trade-off	Participants frame willingness as a trade-off between the bonus and the required time, effort, boredom, frustration, or opportunity cost. Includes mentions of hourly rate, whether \$10 is "worth it," or better alternative uses of time.	"Prolific participants may estimate the time that it will take them to complete the tasks assigned as opposed to completing different surveys offered to them. It is a balance that they must consider when determining whether or not to complete the tasks and the time or effort they are willing to put in."
Task enjoyment	Participants may be more willing because they enjoy the task, find it easy, engaging, fun, challenging, or are more tolerant of boring, repetitive, or mindless work.	"People who enjoy this type of work would be more willing to do more work."
Time availability	Participants may be more willing because they have more free time, fewer schedule constraints, or fewer competing obligations.	"I don't think it's as much about the money as it is on how much free time like person has at the moment."
Misc./Unusable	Response is unclear, too vague to interpret, or explicitly says the respondent does not know.	"I'm not sure."

Table C.2: Open-ended explanations for willingness to complete more tasks for a \$10 bonus

Category	Description	Count	Cumulative %
Financial need	Participants may be willing to do more tasks because they need the money more, including financial hardship, bills, necessities, low income, or reliance on Prolific income.	55 (45.5%)	45.5%
Subjective value of money	Participants may be more willing because they personally value earning \$10 more, are more money-focused, or are more motivated about current or future earnings (getting invited to more studies), even without explicit financial hardship.	45 (37.2%)	71.1%
Time-money trade-off	Participants frame willingness as a tradeoff between the bonus and the required time/ effort/ boredom/ frustration, or opportunity cost. Includes mentions of hourly rate, whether \$10 is “worth it,” or better alternative uses of time.	30 (24.8%)	84.3%
Task enjoyment	Participants may be more willing because they enjoy the task, find it easy, engaging, fun, challenging, or are more tolerant of boring, repetitive, or mindless work.	33 (27.3%)	90.9%
Time availability	Participants may be more willing because they have more free time, fewer schedule constraints, or fewer competing obligations.	25 (20.7%)	92.6%
Misc/ Unusable	Response is unclear, too vague to interpret, or explicitly says the respondent does not know.	9 (7.4%)	100.0%

Notes: The description column uses the exact category-description fed into the LLM for the classification exercise. The Count column reports the number and share of subjects whose answer received each code. Because responses could receive multiple codes, category shares are not mutually exclusive. The cumulative column reports the share of subjects whose answer includes the focal category or any category above it in the table. Responses not fitting in any of the first five categories are categorized as “Misc/ Unusable”.

C.2 A structural model of interpersonal uncertainty and ingroup preferences

Setup. Suppose the representative DM has to distribute M units between two individuals 1 and 2, whose group identity can take one of three values: $G(i) \in \{in, out, \emptyset\}$, where $\emptyset$ means unknown. Suppose the DM believes that the valuations of money received by the two individuals 1 and 2 are distributed as f_1 and f_2 , respectively, and suppose $x_{IU}(f_1, f_2, \gamma)$ is the choice that maximizes the expected (utilitarian) utility (see equation 1) given the two distributions. To parameterize risk aversion, we assume CRRA utility $u(w) = \frac{w^{1-\gamma}}{1-\gamma}$, with γ as the risk aversion parameter. In our experiment, under the assumption that the valuation of money is measured as WTW, f_1 and f_2 are either the degenerate lottery $L_1 = (12, 1)$ (tasks) or the 50-50 lottery $L_2 = (h = 22, 0.5; l = 4, 0.5)$. Under these distributions and under CRRA risk preferences, x_{IU} has the following closed-form solution:

$$x_{IU}(f_1 = (m, 1), f_2 = (h, 0.5; l, 0.5), \gamma) = \frac{M \cdot h \left(\frac{m-l}{h-m}\right)^{\frac{1}{\gamma}} - M \cdot l}{(m-l) + (h-m) \left(\frac{m-l}{h-m}\right)^{\frac{1}{\gamma}}}$$

Finally, we assume that the DM's optimal allocation to individual 1 in observation j is as follows:

$$x_{1j} = \begin{cases} \frac{M}{2} + b + \varepsilon_j & \text{if } f_1 = f_2, G(1) = in, G(2) = out \\ x_{IU}(f_1, f_2, \gamma) + \varepsilon_j & \text{if } f_1 \neq f_2, G(1) = G(2) \\ a_{IU} \cdot x_{IU}(f_1, f_2, \gamma) + a_{ING} \cdot \left(\frac{M}{2} + b\right) + \varepsilon_j & \text{if } f_i \neq f_o, G(1) = in, G(2) = out \end{cases}$$

In the first case of symmetric interpersonal uncertainty, the ingroup preference factor b alone determines the allocation. The normal noise parameter $\varepsilon_j \sim N(0, \sigma^2)$ is i.i.d. across observations. In the second case of symmetric group information, interpersonal uncertainty alone determines the final allocation. When we have a conjunction of the former two cases, the optimal allocation combines the influence of both factors: $a \lesseqgtr 1$ quantifies whether the influence of the corresponding channel shrinks ($a < 1$), stays unchanged ($a = 1$), or expands ($a > 1$) when both factors are present.

Results. We jointly estimate $\gamma, b, a_{IU}, a_{ING}, \sigma$ to maximize the likelihood of the observed data. We estimate a CRRA parameter of $\gamma = 0.374^{40}$ and the extent of pure ingroup preference to be $b = 0.741$. We estimate a weight of $a_{IU} = 0.739$ on interpersonal uncertainty and a weight of $a_{ING} = 0.336$ on ingroup preferences. Thus, when both factors operate simultaneously, the estimated influences of interpersonal uncertainty and ingroup preferences diminish to 74% and 34% of their respective influences when they operated in isolation.

This sub-additive feature inherent in the estimates ($a_{IU}, a_{ING} < 1$) helps one interpret our results from the *social* and *non-social* decisions in light of the reduced-form results from

⁴⁰This implies higher risk aversion toward interpersonal uncertainty than toward monetary risk. For comparison, across 16 studies employing the Gneezy and Potters (1997) investment task over money, which our setup mimics, Crosetto and Filippin (2016) report an average CRRA parameter $\gamma = 0.3$.

the *Ingroup uncertainty* treatment. One might (incorrectly) think that the quantitative similarity between the *social* and *non-social* decisions implies that interpersonal uncertainty is sufficient to explain *all* observed prosocial behavior, which would be at odds not only with the previous literature but also with our results in the *Group information without uncertainty* decision. However, the sub-additivity feature explains that when one compares a treatment where both factors are present to one where only interpersonal uncertainty matters, the influence of interpersonal uncertainty in the latter treatment expands and thus partly compensates for the lack of group preferences.⁴¹

Table C.3: Structural estimation results

	Main model (1)
γ	0.374*** (0.016)
b	0.741*** (0.121)
σ	1.881*** (0.046)
a_{IU}	0.739*** (0.152)
a_{ING}	0.336** (0.133)
LL	-1718
Akaike's IC	3445
Bayesian IC	3469

Notes: γ , b , and σ are the CRRRA parameter, measure of ingroup preferences, and the standard deviation of the noise term ε , respectively. a_{IU} and a_{ING} quantify the importance of interpersonal uncertainty (IU) and ingroup preferences on the optimal allocation choice when both factors are present. See Section C.2 for details. "LL" denotes the maximized log-likelihood, "Akaike's IC" is Akaike's information criterion and "Bayesian IC" the Bayesian information criterion. Significance levels: * $p < 0.1$, ** $p < 0.05$ and *** $p < 0.01$.

C.3 Details on the WTW uncertainty calibration

Through the IU belief and IU WTW elicitation, we have two measures for each social group comparison: first, subjects' sensitivity in the IU rating toward changes in the WTW spread;

⁴¹The unrestricted model fits the data significantly better than several restricted alternatives. Specifically, likelihood-ratio tests reject a model that imposes $\alpha_{IU} + \alpha_{ING} = 1$, as well as models that impose either $\alpha_{IU} = 0$ or $\alpha_{ING} = 0$. These tests account for the additional flexibility of the unrestricted model. Thus, the structural estimates support the interpretation that both interpersonal uncertainty and ingroup preferences contribute to behavior.

second, their perceived IU difference between ingroup and outgroup. We use these measures to construct a WTW spread that is equivalent to their implied IU difference between ingroup and outgroup.

Estimating IU sensitivity. As a first step, we estimate the relationship between subjects' subjective perception of WTW differences and the actual WTW difference. Denote by Y_{WTW} the IU rating subjects give for a given WTW spread. Then, we define X as the difference between the two individuals' WTW values. That is, for the (20, 20) group $X = 0$, for the (18, 24) group $X = 6$, and so on. Accordingly, for each individual i , we observe pairs $\{(X_{ij}, Y_{\text{WTW},ij})\}_{j=1}^4$.

We then estimate for each individual the relationship between WTW spread difference and IU rating by linear regression:

$$Y_{\text{WTW}} = b_0 + b_1 X + \varepsilon. \quad (8)$$

b_1 represents how strongly a subject's IU rating changes with increasing spread. A positive b_1 implies higher IU ratings for larger spreads (as defined by the rating scale direction).

Estimating spreads. The next step is to match this sensitivity to the difference in uncertainty subjects perceive between ingroup and outgroup. For each subject, we elicit their perceived interpersonal uncertainty about ingroup and outgroup for two social groups, politics and hobbies. This gives us four ratings in total, which we denote with Y_G^P . The goal is to find, for each rating Y_G^P , the implied (matching) WTW spread difference X_G^P using the subject-specific regression coefficients. Starting from Eq. (8) and solving for X yields:

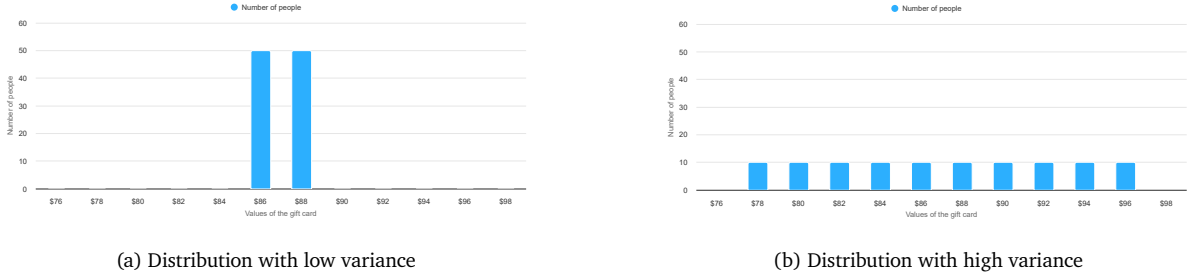
$$X_G^P = \frac{Y_G^P - b_0}{b_1}. \quad (9)$$

This produces four predicted spreads that we subsequently use for the allocation decisions:

$$\{X_G^P\}_{(G,P) \in \text{ingroup/outgroup} \times \text{politics/hobbies}}.$$

Estimation failures. Using this procedure, we are able to estimate spreads for 120 of the 170 subjects. For the remaining 50 subjects, estimation fails because they are either not sensitive to differences in WTW spreads or their responses imply a non-monotonic relationship. For instance, these subjects may report higher perceived uncertainty for a lower spread than for a larger spread. Because we cannot construct spreads for these subjects' allocation decisions, they skip the decision part of the experiment. To assess whether this introduces bias in our quantification exercise, we compare the reported perceived interpersonal uncertainty about the ingroup and the outgroup. Subjects for whom estimation is possible report average ingroup uncertainty of 2.47 and 2.53 in the case of shared political views and hobbies/interests, respectively. By contrast, subjects for whom estimation is not possible report corresponding averages of 2.86 and 2.70. Differences are thus small, and we fail to reject the null of no differences in both instances ($p = 0.30$ and $p = 0.57$, two-sample

Figure D.1: Distributions shown to subjects



Wilcoxon tests). For the outgroup, subjects for whom estimation is possible report average uncertainty of 4.18 and 4.33, while subjects for whom estimation is not possible report 4.68 and 4.26; we again fail to reject the null ($p = 0.26$ and $p = 0.84$, two-sample Wilcoxon tests). Kolmogorov-Smirnov tests comparing the distributions similarly fail to reject the null hypothesis in each instance ($p = 0.14, p = 0.66, p = 0.95, p = 0.92$). These results suggest that our quantification exercise does not systematically misrepresent the distribution of perceived uncertainty.

D Extensions and robustness on measuring interpersonal uncertainty

D.1 Validation of the uncertainty measure

Our self-reported interpersonal uncertainty measure is intended to proxy whether subjects perceive higher interpersonal uncertainty for one group relative to another, as defined in Definition 1 in Section 2. However, it could be the case that subjects instead only report their perceptions of mean differences or concepts unrelated to uncertainty. In this section, we validate that our measure is indeed sensitive to those changes in interpersonal uncertainty captured by our definition. To do so, we provide subjects with two objective WTP distributions, one of which is a mean-preserving spread of the other, and investigate the impact on the answers subjects give to our measure.

Design. At the end of the *Ingroup belief measurement* and *Self/other belief measurement* treatments, we showed subjects two figures. Each figure displayed a frequency distribution of the WTP values of 100 fictitious individuals. In one, 50 individuals had a WTP of \$86 and 50 a WTP of \$88 (low variance distribution). In the other, there were 10 individuals for each of the 10 values between \$78 and \$96 (high variance distribution). See Figure D.1 for the figures shown to subjects. We also provided these values to subjects in text format below the figures. The high variance distribution is a mean-preserving spread of the low variance one, having the same mean but a higher variance. For each figure, subjects were

asked the following about the group displayed in the figure: “Suppose we randomly pick one of the 100 people from this group. How certain are you about how much the randomly chosen person would value the Amazon gift card money?” Subjects could respond on an 11-point Likert scale from *Very uncertain* to *Very certain*, and we re-code the variable so that higher values indicate higher uncertainty. The text and measurement thus closely mirror our self-reported interpersonal uncertainty measure.

Results. We find that subjects report different uncertainty across the two distributions. On average, they report an uncertainty of 3.46 Likert-scale points for the low variance distribution, and an uncertainty of 5.80 points for the high variance distribution, a significant difference ($p < 0.001$, paired Wilcoxon test). On the individual level, 74% of subjects report higher uncertainty for the high variance distribution compared to the low variance distribution, 13% report no difference, and the remaining 13% report more uncertainty for the low variance distribution. Thus, subjects are sensitive to changes in WTP distributions in the expected direction.

D.2 Comparing beliefs elicited with and without prior allocation decision

As described in Section 3.2, in the *Ingroup social* condition, perceived interpersonal uncertainty was elicited after subjects made their *Ingroup social* allocation decision. In *Ingroup IU belief*, it was elicited without a prior choice. This allows us to compare the distribution of beliefs with and without a prior allocation decision.

Results. Table 1 shows that in *IU belief measurement*, subjects report 0.81 points higher uncertainty about outgroup members for hobbies/interests, 0.93 for political views, and 0.66 for religious beliefs (each $p < 0.01$, paired Wilcoxon tests). Aggregating across the three groups, we find that in 37% of cases, subjects report higher uncertainty for the outgroup; in 5% they report higher uncertainty for the ingroup and in the remaining 58% of cases, subjects report no difference. In *Ingroup social*, subjects report 0.76 points higher uncertainty in the case of hobbies/interests, 1.20 in the case of political views, and 0.73 in the case of religious beliefs. In 32% of cases, subjects report higher uncertainty for the outgroup; in 8% they report higher uncertainty for the ingroup and in the remaining 59% of cases, subjects report no difference. Accordingly, in both cases we find similar patterns of differences in interpersonal uncertainty. Importantly, we cannot reject that the distribution of beliefs elicited without a prior allocation decision is the same as the distribution elicited with a prior decision (for each social group $p > 0.10$, Kolmogorov-Smirnov test).

E Extensions and further details on the non-social paradigm and experiments

E.1 Separate results for the gift card and effort environment of the ingroup social/non-social decisions

In Section 3.4.2, we pooled our results across the *Gift card* and *Effort* environments. In this section, we report the results from both environments separately, as preregistered. This allows us to jointly assess whether (i) our results depend on the medium used for allocations and the valuation task (willingness-to-pay for a gift card versus willingness-to-work for bonus payments) and (ii) varying the implementation probability influences choices (paying one per session versus paying one out of every ten subjects).

In the following, we report the results from the *Gift card* environment in the text and supply the results from the *Effort* environment in brackets.

Ingroup social decisions. In the *Ingroup social* decisions, subjects allocate on average \$57.48 (\$58.00), \$71.05 (\$70.03), and \$61.61 (\$59.80) to the ingroup member when they share the same interests/hobbies, the same political views, or the same religious beliefs, respectively. In all cases, we can reject the hypothesis of no ingroup-favoritism ($p < 0.01$, one-sample Wilcoxon tests). In 46% (51%) of the decisions, subjects display ingroup-favoritism by allocating strictly more than 50% to the ingroup. Outgroup-favoritism is found in 8% (3%) of decisions, and in the remaining 46% (45%) of cases, subjects allocate 50/50. In total, 73% (73%) of subjects display ingroup-favoritism in at least one decision.

Ingroup non-social decisions. In both environments, a similar pattern emerges in the *Ingroup non-social* decisions. Here, subjects allocate on average \$56.86 (\$60.02), \$65.02 (\$64.20), and \$60.81 (\$60.17) to the ingroup member's WTP when the ingroup member shares the same interests/hobbies, same political views, and same religious beliefs as the subject. As before, we find behavior resembling ingroup-favoritism in all cases ($p < 0.01$, one-sample Wilcoxon tests). In 61% (54%) of the *Ingroup non-social* decisions, subjects display ingroup-favoritism by allocating strictly more than 50% to the ingroup member's WTP. Outgroup-favoritism is found in 11% (9%) of decisions, and in the remaining 28% (37%), subjects allocate 50/50.

Comparing Ingroup social and non-social. We cannot reject that average ingroup allocations are equal between *Ingroup social* and *non-social* decisions in any case at the 5% level, with p-values of $p = 0.59$ ($p = 0.77$) for hobbies/interests, $p = 0.22$ ($p = 0.06$) for political views, and $p = 0.38$ ($p = 0.68$) for religious beliefs using unpaired Wilcoxon tests.

Within-subject comparison. We also find no substantial difference in the within-subject relationship of *Ingroup social* and *non-social* between the *Gift card* and *Effort* environment.

The correlation is highly significant in both cases, with a correlation coefficient of $r = 0.53$ ($r = 0.68$).

Overall, our results are quantitatively very similar across both environments. This suggests that our results are not specific to either the use of gift cards or effort tasks. Moreover, our results are insensitive to variations in the probability that decisions are implemented with real consequences.

E.2 Within-subject analyses

The results covered in Section 3.4.2 were obtained using a between-subject design, where we only used the first set of decisions each subject faced. In the following, we repeat our analyses using all of the subjects' decisions. In general, our between-subject results replicate well in the within-subject analyses. As in Section 3.4.2, we pool across the *Gift card* and *Effort* environment.

E.2.1 Ingroup versus outgroup decisions

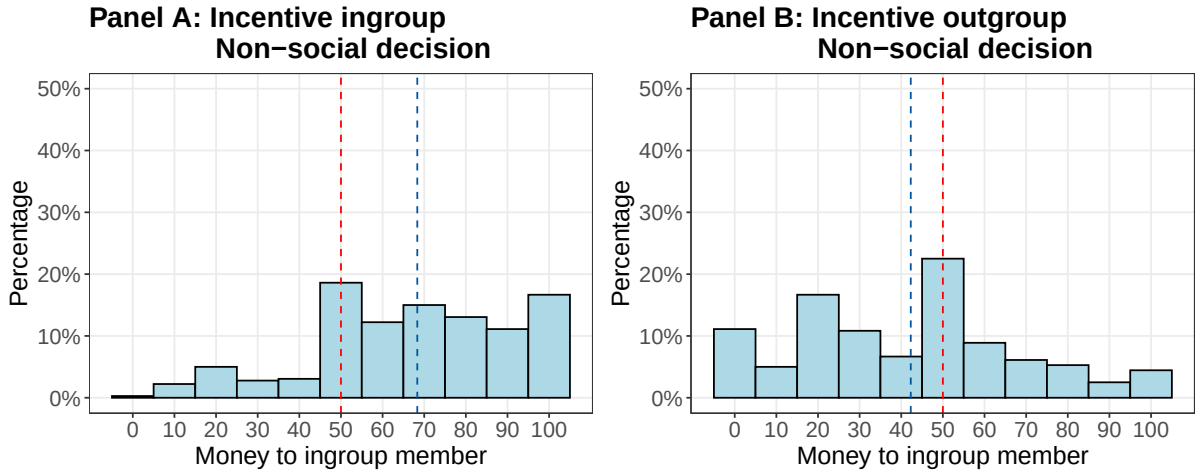
Ingroup social decisions. In the within-subject case, subjects allocate on average \$58.62 if their ingroup members share the same interests/hobbies, \$68.51 if political views are shared, and \$60.92 if religious beliefs are shared. In all three cases, we can reject the hypothesis of no ingroup favoritism ($p < 0.001$, one-sample Wilcoxon tests). In 52% of the decisions, subjects display ingroup favoritism by allocating strictly more than 50% to the ingroup. Outgroup favoritism is found in 6% of decisions, and in the remaining 42%, subjects allocate 50/50. In total, 75% of subjects display ingroup favoritism in at least one decision.

Ingroup non-social decisions. In the *Ingroup non-social* decisions, subjects allocate on average \$59.08 to their ingroup member's WTP when the ingroup is defined by shared interests/hobbies, \$63.65 if political views are shared, and \$60.18 if religious beliefs are shared. Again, we can reject the hypothesis of no ingroup favoritism ($p < 0.001$, one-sample Wilcoxon tests) in all three cases. In 54% of the decisions, subjects display ingroup favoritism by allocating strictly more than 50% to the ingroup member's WTP. Outgroup favoritism is found in 11% of decisions, and in the remaining 35%, subjects allocate 50/50.

E.2.2 Ingroup versus Incentive outgroup robustness

Table E.1 displays the treatment effect of *Incentive outgroup* relative to *Incentive ingroup* separately for the within-subject and between-subject comparisons pooled across the three groups. As displayed, the effect is similar in both the within- and between-subject comparisons. Regarding the within-subject effects in the social groups individually, when the ingroup is incentivized, average allocations to the ingroup WTP increase from \$58.47 to \$67.22 for hobbies/interests ($p < 0.001$, unpaired Wilcoxon tests), from \$64.00 to \$72.57

Figure E.1: Incentive ingroup within-subject



Notes: See Figure 7 for details on the variable definitions. Compared to Figure 7, this figure displays the results using all decisions from subjects in *Incentive Ingroup* and *Incentive Outgroup*. Each panel thus displays $n = 360$ decisions by 120 subjects.

for political views ($p < 0.001$), and from \$58.97 to \$65.22 for religious beliefs ($p = 0.01$) compared to *Ingroup non-social*. Conversely, in *Incentive outgroup*, allocations to the ingroup WTP decrease to \$37.76 for hobbies/interests, to \$46.89 for politics, and to \$42.21 for religious beliefs (all three $p < 0.001$). As in the between-subject comparison, we again see outgroup favoritism in the *Incentive outgroup* decisions. The pooled average is \$42.29, which is significantly different from the even split ($p < 0.001$, one-sample Wilcoxon test). See Figure E.1 for the distributions, which once again show that the shift in average giving is driven by shifts in the distributions.

E.2.3 Self versus others decisions results

Self social decision. In the within-subject case of the *Self social* decision, subjects allocate on average \$68.05 to themselves, thus displaying significant self-regarding behavior relative to the equal split ($p < 0.001$, one-sample Wilcoxon test). In total, 62% of subjects allocate more money to themselves, 9% allocate more to the other person, and 29% implement the 50/50 split.

Self non-social decision. In the *Self non-social* decision, subjects allocate on average \$67.23 to their own WTP, again displaying significant self-regarding behavior ($p < 0.001$, one-sample Wilcoxon tests). In total, 66% of subjects allocate more money to their own WTP, 13% allocate more to the other person's WTP, and 21% implement the 50/50 split.

Comparing Self social and non-social. In the between-subject comparison, we also cannot reject equality of average allocations between *Self social* and *non-social* ($p = 0.69$, paired Wilcoxon tests). Similarly, we cannot reject that the pooled distributions are equal ($p = 0.95$,

Table E.1: Treatment effect of the incentive treatment in the ingroup versus outgroup setting

	Dependent variable:	
	Allocation to ingroup member	
	Within-subject (1)	Between-subject (2)
<i>Outgroup incentive</i>	−26.047*** (3.163)	−25.817*** (3.547)
Constant (<i>Ingroup incentive</i>)	68.333*** (1.937)	67.978*** (2.083)
Subjects	120	120
Observations	720	360
R ²	0.211	0.231

Notes: The table shows OLS estimates. The dependent variable in columns (1) and (2) is the amount subjects allocate to the ingroup member (out of \$100) in the *Incentive outgroup* and *Incentive ingroup* treatments. “Incentive outgroup” is a dummy equal to one if the incentive for the decision gave three times the weight on the outgroup member’s WTP, and equal to zero if the incentive gave three times the weight on the ingroup member’s WTP. In column (1), all decisions are used, in (2) only the first decisions. Standard errors (in parentheses) are clustered at the subject level. Significance levels: * $p < 0.1$, ** $p < 0.05$ and *** $p < 0.01$.

Kolmogorov-Smirnov test).

E.3 Analyzing order effects

A potential concern for the validity of the within-subject results for the *social* and *non-social* decisions (Appendix E.2) is contagion across conditions. As subjects facing the first set of decisions were not aware that a second set would follow, this naturally cannot influence our between-subject analysis presented in the main paper that only uses the first set of decisions. However, subjects may adjust their choice in the subsequent *non-social* decisions to mimic the *social* decisions, potentially biasing the individual-level analyses. Such adjustment could lead to artificially high similarity between the two decisions, and thus artificially high correlations. Because we randomized the order of decisions, we can directly assess this concern by testing for order effects. Overall, we find no evidence that the order influences subjects’ behavior, as we show in the following in detail.

E.3.1 Ingroup versus outgroup decisions

For the *Ingroup non-social* decisions, the pooled average allocations to the ingroup are \$61.50 when elicited before, and \$60.45 when elicited after the *social* decisions. For hobbies/interests, the *Ingroup non-social* averages are \$58.61 and \$59.54 ($p = 0.94$, unpaired

Wilcoxon test), for political views \$64.60 and \$62.73 ($p = 0.49$), and for religious beliefs \$61.29 and \$59.10 ($p = 0.25$). Thus, the averages are invariant to the order. We also cannot reject the null hypothesis that distributions are invariant to the order ($p = 0.78, p = 0.60, p = 0.39$, Kolmogorov-Smirnov tests). Moving to the *Ingroup social* decisions, average allocations are \$63.05 when *Ingroup social* is elicited first, and \$62.31 when elicited after the *non-social* decisions. Again, decisions generally do not differ substantially. For hobbies/interests, the *Ingroup social* averages are \$57.75 and \$59.50 ($p = 0.04$, unpaired Wilcoxon test), for political views \$70.66 and \$66.32 ($p = 0.26$), and for religious beliefs \$60.73 and \$61.12 ($p = 0.21$).

E.3.2 Self versus other decisions

In the case of the *Self non-social* decisions, subjects allocate \$64.12 to themselves when the decision is before the *Self social* decision, and \$70.25 when the decision comes afterward, an insignificant difference ($p = 0.12$, unpaired Wilcoxon test). In the case of the *Self social* decisions, subjects allocate \$69.05 to themselves when the decision is before the *Self non-social* decision, and \$67.02 when the decision comes afterward, again an insignificant difference ($p = 0.61$). In addition, we cannot reject the null that the distributions are invariant to the order either for *Self non-social* ($p = 0.08$, Kolmogorov-Smirnov test) or for *Self social* ($p = 0.54$, Kolmogorov-Smirnov test).

E.4 Details on the robustness treatments of Section 3.4.3

This section provides more details on the results of the robustness treatments that are described in Section 3.4.3.

We start with the between-subject comparison of *Incentive outgroup* versus *Incentive ingroup* by focusing on behavior in the first-assigned choices. We find that subjects respond to changes in the induced utilitarian incentives: Compared to an average giving of \$60.89 to the ingroup members' WTPs across our social groups of the *Ingroup non-social* decisions in the *Gift card* environment, subjects give on average \$42.16 to the ingroup members' WTPs in *Incentive outgroup* and \$67.98 in *Incentive ingroup*, a significant difference (both comparisons $p < 0.001$, unpaired Wilcoxon test). In particular, behavior switches to outgroup favoritism in *Incentive outgroup*, as the average is significantly smaller than the even split ($p < 0.001$, one-sample Wilcoxon test). Regarding the distributions, 54% of subjects in *Incentive outgroup* and 10% in *Incentive ingroup* display outgroup favoritism, while 33% and 72% display ingroup favoritism, respectively. This significant shift in the distributions ($p < 0.01$, Kolmogorov-Smirnov test) shows that the changes in the averages are not driven by a minority of subjects, but by a substantial fraction.

A within-subject analysis reveals that subjects change their behavior in 81% of decisions following the incentive change between the two treatments. Accordingly, in 19% of decisions

subjects are unresponsive to changes in the incentives, indicating inattention or confusion. Comparing the behavior in these situations to the main experiment’s *Ingroup social*, both average ingroup giving (\$60.84 compared to \$63.38) and the fraction of choices displaying ingroup favoritism (34% of decisions compared to 46%) are lower, while 50/50 splits are more frequent (60% compared to 46%).

Next, we turn to the *Incentive minimum* treatment. As predicted, we find that favoritism in either direction is eliminated under the Rawlsian incentive. On average, subjects allocate \$51.31 to the WTP of the individual who shares their interests/hobbies, \$52.85 to the individual who shares their political views, and \$49.77 to the individual who shares their religious beliefs. Hence, the treatment not only significantly reduced ingroup favoritism relative to *Ingroup non-social* (in all three cases $p < 0.01$, unpaired Wilcoxon tests), but also eliminated it altogether, as we can no longer reject equality between average ingroup giving and the 50/50 split ($p = 0.31$, $p = 0.13$, and $p = 0.95$, respectively, one-sample Wilcoxon tests). Moreover, the percentage of decisions that implement exactly a 50/50 split increases from 32% in *Ingroup non-social* to 58% in *Incentive minimum*.

These results provide evidence against limited attention or confusion as drivers of our results in the previous section. Changing a single number in the utilitarian incentive formula reverses the direction of favoritism from ingroup to outgroup favoritism, while moving to Rawlsian incentives eliminates any favoritism. Moreover, in those cases where choices indicate inattention or confusion about the incentives, ingroup favoritism is, if anything, less prevalent relative to the main experiment.

E.5 Self versus others setting incentive treatment

Design. As in the ingroup case, we vary the incentive subjects face when making the *Self non-social* decisions. In *Incentive self*, the weight on the DM’s own WTP is three times as high as the weight on the other individual’s WTP. The DM’s payoff thus becomes:

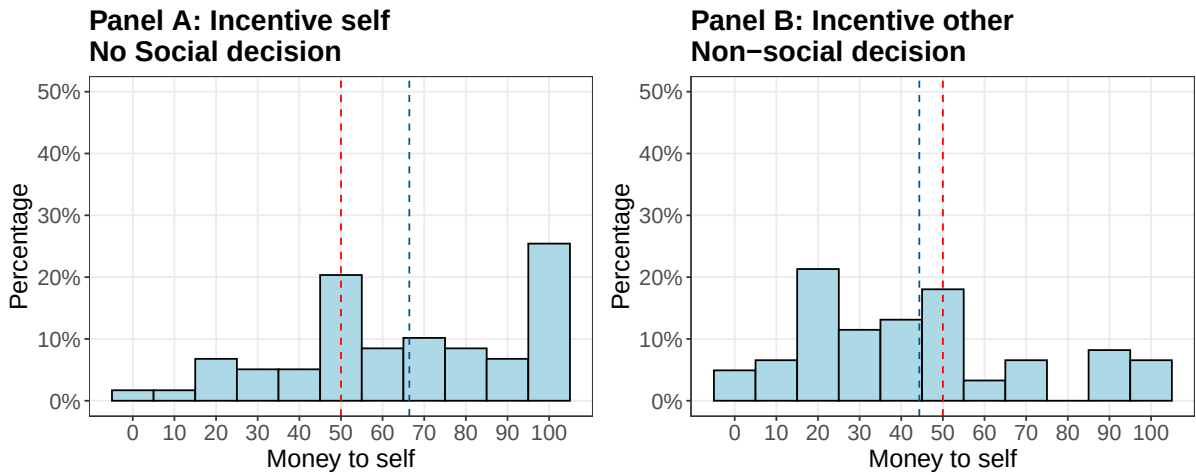
$$\Pi(x_{self}, x_{other}) = 3 \cdot x_{self} \cdot WTP_{self}/100 + x_{other} \cdot WTP_{other}/100$$

In *Incentive other*, we increase the weight put on the other individual’s WTP to three times the DM’s WTP:

$$\Pi(x_{self}, x_{other}) = x_{self} \cdot WTP_{self}/100 + 3 \cdot x_{other} \cdot WTP_{other}/100$$

Results. Inducing these incentives changes people’s behavior in the *non-social* decision. See Table E.2 for the within-subject and between-subject treatment effect. In both cases, the change in incentives leads to a significant change in the amount participants allocate to their own WTP. In the within-subject comparison, participants allocate \$19.50 less to their own WTP when incentives are higher for the other participant; in the between-subject comparison, they allocate \$22.08 less. Figure E.2 displays the distributions in the between-subject case. The fraction of subjects allocating more than 50% of the endowment to their

Figure E.2: Self versus other incentive



Notes: Histogram of *Incentive self* (Panel A) and *Incentive other* (Panel B) decisions. The x-axis denotes the amount of gift card money (out of \$100) that subjects allocate to themselves instead of another individual. Subjects' incentive is to maximize the weighted sum of their own and another individual's WTP. In Panel A, subjects' own WTP receives three times the weight; in Panel B, the other individual's WTP receives three times the weight. The red dotted line denotes the even split benchmark, the blue dotted line the average allocation. For both panels, the binwidth is 10. Only the first decision is used for each subject. Panel A displays $n = 59$ decisions by 59 subjects; Panel B displays $n = 61$ decisions by 61 subjects.

own WTP increases from 33% in *Incentive other* to 63% in *Incentive self*, while the fraction of subjects allocating more money to the other participant decreases from 50% to 22%. The distributions are significantly different from each other ($p < 0.001$, Kolmogorov-Smirnov test).

The within-subject comparison shows that 83% of subjects change their allocation behavior between *Incentive self* and *Incentive other*. Among those 17% of subjects who are unresponsive to the incentive change, 33% allocate more to themselves, a substantially lower fraction than the 58% in the main *Self non-social* case. In total, 38% choose the equal split, and 29% allocate more to the other participant. Taking the behavior of these unresponsive subjects as indicative of inattention or confusion, it appears that such factors are associated with subjects allocating less to themselves. This result thus provides suggestive evidence that our replication of significantly more self-giving in *Self social* using the *Self non-social* decisions is not driven by inattentive or confused subjects.

F Experimental instructions and research transparency

Instructions. The instructions of all experiments can be found in the following Open Science Framework (OSF) repository:

https://osf.io/tcp3d/?view_only=27899531990048e4b608d64b13528236

Preregistration. All experiments covered in the paper were preregistered at aspredicted.org. The preregistrations include details on the experimental design, the planned sample size,

Table E.2: Treatment effect of the incentive treatment in the self versus other setting

	<i>Dependent variable:</i>	
	Allocation to self	
	Within-subject (1)	Between-subject (2)
<i>Incentive other</i>	−19.500*** (3.468)	−22.079*** (5.045)
Constant (<i>Incentive self</i>)	65.625*** (2.563)	66.424*** (3.592)
Subjects	120	120
Observations	240	120
R ²	0.108	0.140

Notes: The table shows OLS estimates. The dependent variable is the amount subjects allocate to themselves (out of \$100) in the *Incentive other* and *Incentive self* treatments. “Incentive other” is an indicator equal to one if the incentive for the decision gave three times the weight on the other person’s WTP, and zero if the incentive gave three times the weight on the subject’s own WTP. In column (1), all decisions are used, in (2) only the first decisions. Standard errors (in parentheses) are clustered at the subject level. Significance levels: *p<0.1, **p<0.05 and ***p<0.01.

exclusion criteria, hypotheses, and the main analyses. Table F.1 provides an overview of the treatments and links to the respective preregistrations.

Mapping of pre-registration and implementation. Our experimental implementation closely followed the pre-registration. In particular, we implemented the experimental design and sample size exactly as specified in the pre-registration. Similarly, we employed the exclusion criteria as pre-registered: we specified to exclude any subject who did not complete the experiment. This led to the exclusion of 15 subjects in the *Synthetic group* treatment, 4 in *Belief measurement*, 15 in *Ingroup uncertainty calibration*, 22 in the *Ingroup social* and *Ingroup non-social* treatments, 13 in *Ingroup effort social* and *Ingroup effort non-social*, 28 in *Incentive ingroup* and *Incentive outgroup*, 6 in *Ingroup minimum*, 22 in *Self social* and *Self non-social*, 23 in *Incentive self* and *Incentive other*, and 6 in the *Contact* and *Identifiable victim* interventions. The sample sizes reported in Table F.1 are the final sample sizes used in all analyses of the paper after excluding the previously mentioned numbers of subjects.

Deviations from the pre-registration. The *Gift card* and *Effort* environments of the *Ingroup social* and *Ingroup non-social* treatments were pre-registered and run separately (the *Effort* environment as a robustness check for the *Gift card* environment). In the main text, we pool both environments together. In Appendix Section E.1, we report the separate results, as pre-registered. The mediation analysis reported in Section 4.2.1 was not pre-registered.

Table F.1: Overview of treatments

Label	<i>N</i>	Date	Covered in	Link to preregistration
<i>Synthetic group paradigm</i>	120	May 2024	Section 3.1	https://aspredicted.org/rzkc-d3gh.pdf
<i>Belief measurement</i>	120	June 2024	Sections 3.2	https://aspredicted.org/wxtg-7b5m.pdf
<i>Ingroup uncertainty calibration</i>	120	March 2026	Section 3.3	https://aspredicted.org/6mv2e8.pdf
<i>Ingroup social/non-social paradigm: Ingroup social & Ingroup non-social Gift card environment</i>	119	January 2024	Section 3.4	https://aspredicted.org/9cpb-y7dh.pdf
<i>Ingroup social/non-social paradigm: Ingroup effort social & Ingroup effort non-social Effort environment</i>	121	May 2024	Section 3.4	https://aspredicted.org/rzkc-d3gh.pdf
<i>Ingroup social/non-social paradigm: Incentive ingroup & Incentive out-group</i>	120	January 2024	Section 3.4.3	https://aspredicted.org/9cpb-y7dh.pdf
<i>Ingroup social/non-social paradigm: Ingroup minimum</i>	62	February 2024	Section 3.4.3	https://aspredicted.org/dshn-fznj.pdf
<i>Self versus other social/non-social paradigm: Self social & Self non-social</i>	120	February 2024	Section 4.1	https://aspredicted.org/8sb6-wz9h.pdf
<i>Self versus other social/non-social paradigm: Incentive self & Incentive other</i>	120	February 2024	App. Section E.5	https://aspredicted.org/8sb6-wz9h.pdf
<i>Behavioral interventions paradigm Contact and Identifiable victim interventions</i>	602	January 2025	Section 4.2	https://aspredicted.org/37tq-7skc.pdf

Omitted treatments. The pre-registrations for the *Ingroup social* and *Ingroup non-social Gift card* treatments as well as the *Self social* and *Self non-social* treatments contain another set of treatments labeled *Group info* and *Self info*. These treatments are not part of this paper and their results are available upon request because the design is superseded by the *Ingroup uncertainty* experiment. The omitted treatments show that providing subjects with information on the WTP of the recipients significantly changes their allocation behavior both in *Ingroup social* and *Self social*. However, in contrast to the *Ingroup uncertainty* experiment, this information manipulation does not directly manipulate interpersonal uncertainty and is potentially confounded by experimenter demand effects.

Omitted experiments. As part of our initial data collection, we pre-registered and implemented another experiment that is omitted in this version. This experiment used the social/non-social paradigm and applied it to a dictator game with a taking option. It was conducted in February 2024, had 123 subjects and featured two treatments: *Self taking social* and *Self taking non-social*. The design and results were presented in an earlier version of the paper, for instance in the following version: <https://docs.iza.org/dp17708.pdf>. The experiment's pre-registration can be accessed under the following link: <https://aspredicted.org/25wz-bw8x.pdf>